QUANTITATIVE APTITUDE

CAT 2026 | Practice Questions

Questions, Answer Keys and Worked Solutions

Clean study edition

Questions

Instructions
For the following questions answer them individually

Question 1

Superman and Flash race around the Earth on a 100-unit-long track. Superman flies at a constant speed throughout the race, while Flash's speed over the ocean is 20% lower than his speed over land. The ocean constitutes 60% of the total track. On the land portion, Superman takes half the time Flash takes. If Flash takes 39 seconds longer than Superman to complete the entire race, find the sum of their speeds on land, in units per second.

A 15

B 10

C $\frac{10}{3}$

D 5

Question 2

Raman is travelling from city A to city B. He covers 30% of the total distance at 40% of his usual speed, 20% of the total distance at 10% of his usual speed and the remaining distance at 50% of his usual speed. His average speed for the journey is 80 km/hr. If he had travelled the complete distance at his usual speed, he would have covered the whole distance in 8 hours. What is the distance between A and B (in km)?

Question 3

Workers A and B together can complete a commercial project in 12 days. Workers B and C together can complete the exact same project in 20 days. A starts working on the project alone and leaves after 3 days. B then takes over and works for 8 days before leaving. Finally, C takes over and finishes the remaining work in exactly 35 days. If the total contract value for the project is Rs. 1,44,000, and it is distributed based on the exact proportion of work done by each individual, what is the difference (in Rs.) between the final share of money of A and B?

A 14400

B 16800

C 18000

D 15600

Question 4

Vessel A contains milk and water in the ratio 4:1, and Vessel B contains milk and water in the ratio 2:1. A specific volume from Vessel A and a specific volume from Vessel B are mixed to form a 60-liter mixture. From this new mixture, 15 liters are drawn off and completely replaced by pure milk. If the final ratio of milk to water in the resulting mixture is exactly 4:1, what was the initial volume taken from Vessel A (in liters)?

Question 5

A man, a woman, and a child can complete a piece of work together in 20 days. A man and a woman can complete the same work in 25 days, while a woman and a child can complete it in 40 days. How many men are required to complete the entire work in 2 days if only men are allowed to work and each man has the same efficiency?

Question 6

A train crosses a man running at a speed of 5 m/s in the same direction as the train in 15 seconds. The same train crosses a second train, 200 m long and moving in the opposite direction at 20 m/s, in 9 seconds. Find the length of the first train (in m).

- **A** 90
- **B** 62.5
- **C** 80.25
- **D** 78.33

Question 7

A, B, and C together can complete a piece of work in more than 6 days but less than 15 days. If the efficiency of A is half of the combined efficiency of B and C, and the efficiency of C is half the efficiency of B, then find the minimum number of integral days taken by B to complete the work.

Question 8

For a milk seller, the cost price of 10 litres of pure milk equals the selling price of 7 litres of a milk-water mixture he sells to his buyers. If the mixture he sells contains 30% water, which costs nothing, find his net profit percentage.

- **A** $98.54\ \%$
- **B** $104.08\ \%$
- **C** $109.28\ \%$
- **D** $105.12\ \%$

Question 9

Two mixtures contain X and Z in the ratios $11:7$ and $5:1$, respectively. When the two mixtures are combined in the ratio $a:5$, the resulting mixture contains X and Z in the ratio $a:1$. This mixture is then again mixed with the 2nd initial mixture in the ratio $a:a+1$. What will be the approximate concentration of X in this final mixture?

- **A** 75%
- **B** 78%
- **C** 80%
- **D** 82%

Question 10

$ab + bc = 119$, and $a - c = 5$. If a, b, and c are positive integers, then find the maximum value of abc.

Question 11

A milkman splits pure milk into containers A and B in a $3 : 5$ ratio. He replaces 10% of A's mixture with water three consecutive times, and 30% of B's mixture with water three consecutive times. If he sells the combined final solutions at his cost price of pure milk, what is his overall profit percentage, considering that water is free of cost?

- **A** $110.11\ \%$
- **B** $130.56\ \%$
- **C** $121.66\ \%$
- **D** $105.02\ \%$

Question 12

$$\frac{2}{a} + \frac{3}{4b} = \frac{1}{4}$$

Find the difference between the maximum and minimum possible values of $(a + b)$, given that a and b are non-negative integers.

Question 13

In a triangle XYZ, the lengths of XY, YZ, and ZX are $16, 20, 18$ cm, respectively. P is the midpoint of YZ, and PQ is the median to XZ in triangle XPZ. Find the length of the median from Q to XP in triangle XPQ.

- **A** 5 cm
- **B** $5\sqrt{2}$ cm
- **C** 10 cm
- **D** $\frac{20}{\sqrt{3}}$ cm

Question 14

ABC is an equilateral triangle inscribed in a circle and P is a point on the minor arc BC (the arc not containing A). If PB = 3 cm and PC = 5 cm, what is the value of $AP^2 + BC^2$ (in sq. cm)?

- **A** 105
- **B** 113
- **C** 121

D 128

Question 15

ABCD is a trapezium, with AB || CD. The diagonals of the trapezium intersect at F. If the Area of AFB is 9 sq. cm, the length of AB = 3 cm, and the length of CD = 4 cm, find the height of the trapezium.

A 14

B 15

C 7

D 9

Question 16

A triangle ABC has an area of $25\ \mathrm{cm}^2$. A point D on BC divides it in the ratio $BD:DC=2:3$. Next, a point E on AB is chosen such that $BE:EA=2:3$. Find the difference in the areas of triangles DEB and ADC(in cm^2).

Question 17

In a rectangle ABCD with AB = 8cm and AD = 4 cm, point F lies on side DC and point E lies on side BC such that $\angle AFE=90^\circ$. If the ratio of the area of triangle ADF to the area of triangle FCE = 4:9, find the area of triangle AFE

A $15\mathrm{cm}^2$

B $\sqrt{450}\mathrm{cm}^2$

C $30\mathrm{cm}^2$

D $\sqrt{150}\mathrm{cm}^2$

Question 18

In a race of 600 metres, P gives Q a head start of 50 metres and still wins by 10 seconds. In the same race, Q won against R by 6 seconds. If P and R participated in the race of 1000 metres, and the speed of Q is 25 m/sec, then find the time by which R will lose.

A 20 seconds

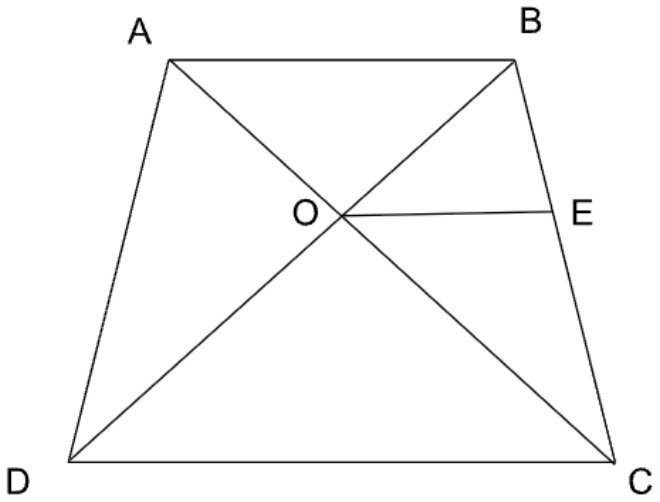
B 25 seconds

C 30 seconds

D 35 seconds

Question 19

In trapezium $ABCD$, with AB and CD being the parallel sides such that $AB : CD = 3 : 5$, and line OE is parallel to the side DC. Find the ratio of the areas of triangle BOE to triangle BOC



A $2 : 5$

B $3 : 7$

C $5 : 11$

D $3 : 8$

Question 20

A and B together can complete a job in 12 days, while B and C together can complete it in 15 days. A works alone for 4 days, after which B joins A. After working with B for 3 days, A's efficiency increases by 25%. A and B then work together for another 4 days, after which B leaves and C joins A, following which A's efficiency decreases by 20%. A and C together complete the remaining work in 2 days. If A and C work on alternate days with C starting on the first day, how many days will they take to complete the job?

Question 21

Four natural numbers a, b, c, and d have an average of 37. When a, b, c, and d are increased by 40%, 30%, 20%, and 10% respectively, their average becomes 44.15. Among all such quadruples (a, b, c, d) of natural numbers, consider the one for which the value of a^3b^2c is the largest possible. Find the value of d in that quadruple.

A 79

B 69

C 74

D 64

Question 22

A person walks up an upward-moving escalator and reaches the top in 15 seconds. If he runs up the same escalator, he reaches the top in 10 seconds. If his walking speed and running speed are in the ratio $4:9$, find the time taken (in sec) by the escalator to carry him to the top when he is stationary.

Question 23

Manoj and Rushi are taking part in a hop race, where the distances covered by them in each hop are in the ratio 3:5. Rushi completes the race in 60 hops. If both of them take 1 second to complete each hop, after how many hops of Manoj should Rushi start hopping so that they finish the race at the same time?

Question 24

Amal and Bimal are running on a circular track in the same direction, with their speeds in the ratio 4:7. Chetan joins them by running in the opposite direction at twice Amal's speed. If all three begin their runs simultaneously from the same point, at how many distinct points on the track will any of them meet each other?

A 24

B 18

C 15

D 21

Question 25

The speeds of A, B and C are integers in the ratio 2 : 3 : 5. In a 100 m race, B starts first. A and C start together 5 seconds later, with A starting 10 m ahead of the starting line. If C wins the race, then find the maximum distance by which B beats A.

A $\frac{100}{3}$ m

B $\frac{110}{3}$ m

C $\frac{130}{3}$ m

D $\frac{140}{3}$ m

Question 26

A container initially contains 72 L of milk. In two successive replacements, 24 L and x L of the mixture are replaced with water, respectively. If the final concentration of milk in the solution is 33.33%, find the value of x.

A 18 L

B 24 L

C 9 L

D 36 L

Question 27

64 litres of milk is initially present in a container. A milkman replaces 16 litres of the liquid in the container with water. He repeats this procedure until the amount of milk remaining in the container is 27 litres. How many times in total does he perform this replacement?

A 3

B 6

C 2

D 4

Question 28

Three solutions contain milk in the percentages $20\%, 36\%,$ and 40% respectively. In what ratio should the three solutions be mixed to obtain a solution containing 25% milk?

A $14:5:1$

B $14:3:1$

C $14:5:3$

D $12:5:1$

Question 29

Find the sum of all possible real values of x if $(2-\sqrt{3})^{x^2-2x-2}+(2+\sqrt{3})^{x^2-2x-2}=4$

Question 30

Two cars start from point A at the same time. The faster car reaches the point B after 1 hour 2 minutes and 30 seconds. After reaching the destination, the car turns around and meets the slower car after 20 minutes and 50 seconds. If the speed of the slower car is 24 kmph, find the distance between A and B.

A 50 km

B 48 km

C 96 km

D 75 km

Question 31

A person bought some mangoes, oranges, and grapes, accounting for a total of 145 fruits. If they sold $\frac{2}{3}$ of the grapes, half of the oranges and $\frac{1}{5}$ of the mangoes and were left with 66 fruits, find the initial number of mangoes if the initial number of grapes and oranges are distinct multiples of 25.

A 20

B 75

C 50

D 100

Question 32

Find the value of x for which $\frac{(x^2+1)^2+9x^2}{x(x^2+1)}$ attains its minima given $x>1$.

A $\frac{3+\sqrt{5}}{2}$

B $\frac{3-\sqrt{5}}{2}$

C $\frac{5-\sqrt{5}}{2}$

D $\frac{5+\sqrt{5}}{2}$

Question 33

In a triangle, length of one of the sides is 4 cm and the angle opposite to this side measures 30°. The length of the circumradius of this triangle is

A 16 cm

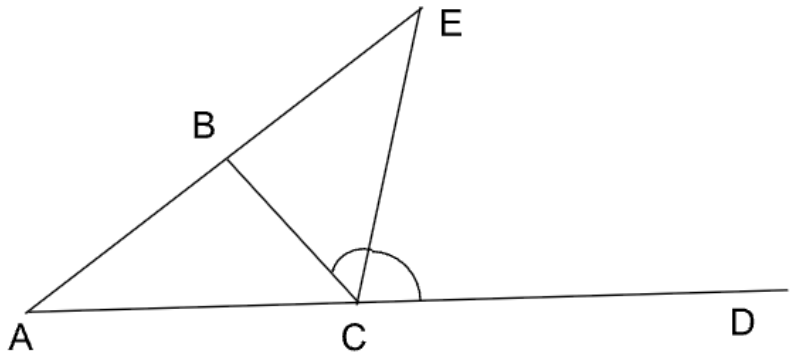
B 2 cm

C 8 cm

D 4 cm

Question 34

In the given figure, CE is the angular bisector of $\angle BCD$. If $\frac{AB}{BE}=\frac{2}{3}$ and $AC=5$ cm, the length of BC is



A 7.5 cm

B 6 cm

C 3 cm

D 4.5 cm

Question 35

An n sided polygon has 350 diagonals. Find the value of n

A 27

B 25

C 32

D 28

Question 36

The number of isosceles triangles with integral side lengths and a perimeter of 20 cm is

A 6

B 5

C 4

D 8

Question 37

Alloy 1 contains Copper, Zinc and Tin in the ratio 2:3:5. Alloy 2 contains Copper, Zinc and Tin in the ratio 4:1:5. Alloy 3 contains Copper, Zinc and Tin in the ratio 3:4:3. An alloy maker mixes alloy 1, alloy 2, and alloy 3 in the ratio 4:5:6. If the amount of Tin present in the mixture is 252 grams, what is the amount of Copper(in grams) present in the mixture?

A 126

B 207

C 184

D 189

Question 38

A cricket batsman goes to an improvement camp. He plays eight matches there. His scores always increases from match 1 to match 8. The average of his scores from match 1 to match 8 is 21.5, whereas the average of his scores in the first six matches, and in the last five matches is 19.833, and 23.6 respectively. What is the minimum he scored in match 5?

A 19

B 20

C 21

D 22

Question 39

The pair of linear equations $ax + 6y = 13$ and $7x + by = 7$ has no solution. Let k be the number of positive integral factors of the product ab, then find k.

A 2

B 4

C 8

D 16

Question 40

A train of length 1050 m is moving at a speed of 10 m/s. A car starts from the back end (caboose) of the train and travels along the train at a speed of 25 m/s (with respect to the ground). It reaches the front end (engine) of the train, immediately turns around, and travels back to the caboose at the same 25 m/s speed (with respect to the ground). Find the distance travelled by the train during the entire journey of the car.

A 1000 m

B 1200 m

C 1250 m

D 2000 m

Question 41

A person travels a certain distance at his usual speed. If he were to travel 6 km/h slower than his usual speed, he would take 4 hours more to complete the journey. If he were to travel 4 km/h faster than his usual speed, he would take 1 hour less than the usual time. What is the person's usual speed?

A 12 km/hr

B 8 km/hr

C 16 km/hr

D 10 km/hr

Question 42

A bus travels at an average speed of 60 km/h when it is moving continuously without any stoppages. However, including stoppages, its average speed reduces to 45 km/h. For every hour that the bus is running, how much time does it spend at stoppages?

A 15

B 20

C 25

D 10

Question 43

Consider a triangle ABC. AM is a median drawn to the side BC. The length of AB=4 cm, AC=6 cm, AM=3 cm. What is the length of BM?

A 3 cm

B 4 cm

C 2 cm

D $\sqrt{17}$ cm

Question 44

The three sides of a triangle are 13 cm, 14 cm, and 15 cm. What is the inradius of the triangle (in cm)?

A 4

B 6

C 8

D 10

Question 45

In △ABC, line DE is drawn parallel to BC, where D lies on AB, and E lies on AC. If AD: DB = 1: 2 and the area of △ADE = 9 sq. cm, find the area of trapezium BCED(in sq. cm).

A 31.5

B 63

C 81

D 72

Question 46

Find the area enclosed by the equations: $x^2 + y^2 \leq 25, y \geq x$ and $y \geq -x$.

A $\frac{5\pi}{4}$

B $\frac{25\pi}{4}$

C $\frac{20\pi}{4}$

D $\frac{\pi}{4}$

Question 47

Find the absolute difference between the sum of the first 10 even numbers and the sum of the first 10 odd numbers.

Question 48

Determine the product of x and y, given that the equation $9x + 2y = 9 - 3xy$ has non-negative integer solutions.

A 5

B 0

C 3

D 2

Question 49

Determine the value of $x + y$ given that the equation $4xy + 4x + y = 26$ has positive integer solutions.

A 6

B 8

C 4

D 2

Question 50

The speed of a boat in still water is 10 km/hr, and the speed of the river current is 2 km/hr. The boat travels a certain distance downstream in 2 hours. What is the time taken by the boat to cover double the previous distance while travelling upstream?

A 4 hours

B 5 hours

C 6 hours

D 8 hours

Question 51

A person walks up a moving escalator and reaches the top in 20 seconds. If he stands still on the same escalator, he is carried to the top in 60 seconds. How long will he take to reach the top if he walks up at the same speed while the escalator moves downward at the same speed as before?

A 30 seconds

B 60 seconds

C 40 seconds

D 45 seconds

Question 52

Ram and Shyam run between A and B with their speeds in the ratio 1:7, with Ram starting from A and Shyam from B. They run towards each other and meet for the first time after five minutes. If the distance between A and B was 20% larger, the point they would have met would've been 840 meters from B. Find the time taken (in seconds) for them to meet if the speed of Shyam increases to twice what it is now.

Question 53

The monetary value of a rare gemstone varies directly as the square of its weight. Unfortunately, a large gemstone is accidentally dropped and breaks into three distinct pieces whose weights are in the ratio 1 : 2 : 3. If the value of the original, unbroken gemstone was Rs. 72,000, what is the total financial loss incurred due to the breakage?

A Rs. 38,000

B Rs. 44,000

C Rs. 48,000

D Rs. 52,000

Question 54

The agricultural yield (Y) of a particular crop depends directly on the product of the amount of fertilizer applied (F) and the square root of the annual rainfall (√R). In a year when a farmer applies 50 kg of fertilizer and receives 64 cm of rainfall, the crop yield is 2400 kg. If the rainfall in the following year is expected to be 81 cm and the farmer aims for a yield of 3240 kg, how much fertilizer should be applied?

A 45 kg

B 54 kg

C 60 kg

D 72 kg

Question 55

Part of the payment of a tourism plan for a group is fixed and the rest varies with the square of the number of tourists purchasing the plan in the group. A group can at most have 10 tourists. 8 friends plan to purchase the plan and are quoted a price of Rs. 5,16,000. If 4 more friends join and 2 of them join the earlier group, the total quoted price becomes Rs. 8,76,000. What is the minimum amount the 12 friends can pay if they are divided into two groups?

A Rs. 6,50,000

B Rs. 6,68,000

C Rs. 5,72,000

D Rs. 6,25,000

Question 56

Let $f(x) = x^2 + px + q$, where $p, q \in R$. Suppose that for every real value of k, the equation $f(x) = kx$ has real solutions. Then which of the following must be true?

A $q \leq 0$

B $p = 0$

C $p^2 \leq 4q$

D $q \geq 0$

Question 57

Two trains, A and B, running at a speed of 108 kmph and 90 kmph, respectively, cross a tunnel. The length of A and B is 200 metres and 250 metres, respectively. The time it takes for the rear end of train A to enter the tunnel, after its engine has just exited the tunnel is x seconds. Similarly, this time for train B is y seconds. If $x + y = 5\frac{2}{3}$ seconds, find the length of the tunnel.

A 100 metres

B 150 metres

C 200 metres

D 250 metres

Question 58

A wholesaler purchases a certain quantity of mangoes at a fixed price per unit and plans to sell them at the same price. He later discovers that 15% of the mangoes are rotten and increases the selling price of the remaining mangoes by Rs. 10 per unit, thereby earning a 19% profit. If instead 20% of the mangoes were rotten and he increased the selling price of the remaining mangoes by Rs. 15 per unit, what would be his percentage profit?

A 17%

B 22%

C 24%

D 28%

Question 59

If α and β are the roots of the quadratic equation $x^2 - x - 1 = 0$, find the value of $\alpha^4 + \beta^4$.

Question 60

Find the number of non-negative integers that satisfy the inequality $\left|\left|x^2 - 5x + 2\right| + 7\right| \leq 11$

Question 61

Find the total number of all the integral solutions that are possible for the inequality $|X + 1||X + 7| < 12$

A 7

B 8

C 9

D 10

Question 62

Find the number of integral values of x that satisfy the inequality $\frac{|x - 2|\left(x^2 - 11x + 28\right)}{(x^2 - 2x - 35)} < 0.$

A 6

B 8

C 7

D 9

Question 63

In a triangle PQR, PQ = 20 cm, QR = 24 cm, and median PS from vertex P on side QR is of length $8\sqrt{7}$ cm. Find the length of the median from vertex Q on side PR.

A $2\sqrt{73}$

B $2\sqrt{87}$

C $4\sqrt{73}$

D $3\sqrt{87}$

Question 64

Ajay and Vijay start moving from the same point in opposite directions around a circle. Their speeds are in the ratio 2 : 3 respectively. When they meet for the 9th time, the shortest distance between the starting point and the meeting point is 200 m. Find the distance travelled by Ajay when they meet for the 7th time.

A 2100

B 1400

C 2800

D 700

Question 65

The profit made by selling a block of wood is proportional to the surface area of the wood. A seller earns Rs. 10,000 when he sells a block of wood with dimensions 2m x 2m x 2m. If the block is broken into five pieces, one with dimensions 2m x 2m x 1m and another four cubes with sides 1m each. How much does the seller earn now by selling the five pieces?

A Rs. 20000

B Rs. 13333

C Rs. 15000

D Rs. 16667

Question 66

A task that originally required 55 men working for 6 days to complete is now constrained by a labor shortage, with only women and children available. A woman is 20% less efficient than a man, and a child is 50% less efficient than a man. At the start of each day, one worker (either a woman or a child) is hired randomly and continues working until the task is finished.

Let S represent the set of all possible integer values for the number of days required to complete the task under these conditions. What is the sum of all elements in set S?

A 236

B 248

C 260

D 292

Question 67

A sum of money becomes ₹13,380 after 3 years and ₹20,070 after 6 years on compound Interest. Find the sum?

A ₹8,120

B ₹8,420

C ₹8,520

D ₹8,920

Question 68

The simple interest accrued on a sum of certain principal in 8 years at the rate of 13% per year is ₹6500. What would be the compound interest accrued on that principal at the rate of 8% per year in 2 years?

A ₹1040

B ₹1020

C ₹1060

D ₹1200

Question 69

Himanshu and Nikhil walk along a circular track, simultaneously starting from the same point. If they walk in the same direction, then in 45 minutes, Nikhil completes 3 more rounds than Himanshu. If they walk in opposite directions, they meet for the first time exactly after 5 minutes. What is the ratio of Himanshu and Nikhil's walking speeds?

A 1:4

B 1:3

C 2:3

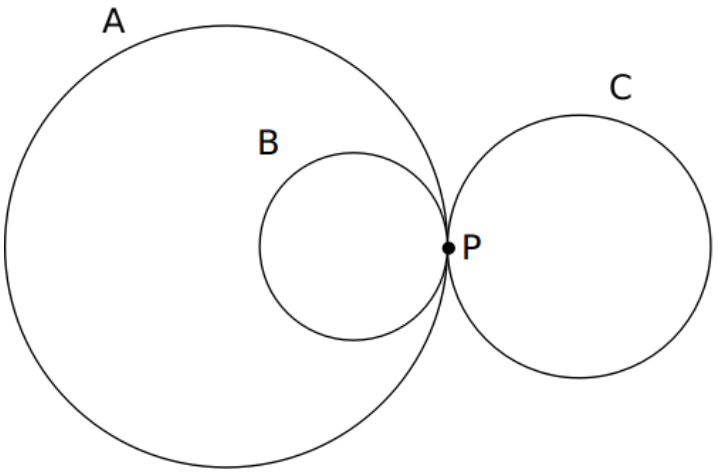
D 1:2

Question 70

Aman and Kushal co-authored a thesis paper consisting of 288 pages. Aman worked thrice the time Kushal worked on the paper, but Kushal wrote 4 pages more than Aman per hour. Both worked on the paper for a total of 36 hours. If the speeds of Aman and Kushal were interchanged, how much approximate time (in hours) would Kushal take to write the thesis paper alone? (Approximate your answer to the nearest integer)

Question 71

Three circular tracks A, B and C meet a point P. The diameter of A, B and C are 18 metres, 7 metres and 10 metres respectively. X, Y and Z start running from point P in tracks A, B and C respectively such that X and Z run in the clockwise direction and Y run in the anti-clockwise direction. All three meet at P after each round at the same time. If the speed of X is 144 km/h, and the absolute difference between the speeds of Y and Z (in m/s) is r, find the value of 12r.



Question 72

A merchant buys 2L pure milk and mixes 3L water with it. The cost of water is 1/100th times the cost of milk. If the merchant sells the mixture at the price of pure milk, find the profit% in the transaction.

A 46%

B 146%

C 64%

D 164%

Question 73

Arjun brought his two cousins to the playground whose names were Atul and Arun. When his friends asked their ages he told them that the sum of the age of Atul and inverse of the age of Arun when multiplied by sum of the age of Arun and inverse of the age of Atul gives you 49/6. The ages of both the cousins were natural numbers

greater than 1. What are the ages of the two cousins?

A 2,3

B 3,4

C 2,4

D 2,6

Question 74

A shopkeeper markups a price of an item by a certain percent. Later on, while selling he gives a discount % which is half the markup percent. If he makes an overall profit of 8%, what can be the markup percent?

A 20%

B 30%

C 32%

D 44%

Question 75

10 years from now ratio of age of Ram and Shyam will be 6:7. 15 year from now their age difference will be 5 years. What is the current age of Ram

A 15

B 20

C 25

D 30

Question 76

Hari works at a mango plantation farm. Each mango tree at the farm contains 240 mangoes on an average. If there are n trees in the garden and he plucked x mangoes from each tree, 5200 mangoes are left in total at the farm. Had he plucked 20 more mangoes from each tree, 4400 mangoes would remain in total. What is the value of x?

A 120

B 90

C 70

D 110

Question 77

A person buys 18 local tickets for Rs. 110. Each first class ticket costs Rs. 10 and each second class ticket costs Rs. 3. What will another lot of 18 tickets in which the number of first class and second class tickets are interchanged cost?

A 112

B 118

C 121

D 124

Question 78

A man purchased 40 fruits: apples and oranges for Rs. 17. Had he purchased as many oranges as apples and as many apples as oranges, he would have paid Rs. 15/-. Find the cost of one pair of an apple and an orange.

A 70 paise

B 60 paise

C 80 paise

D 1 rupee

Question 79

In a factory, each day the expected number of accidents is related to the number of overtime hour by a linear equation. Suppose that on one day there were 1000 overtime hours logged and 8 accidents reported and on another day there were 400 overtime hours logged and 5 accidents. What is the expected number of accidents when no overtime hours are logged?

A 2

B 3

C 4

D 5

Question 80

Nisha went to buy three types of stationery products, each of them were priced at Rs. 5, Rs, 2 and Rs. 1 respectively. She purchased all three types of products in more than one quantity and gave Rs. 20 to the shopkeeper. Since the shopkeeper had no change with him/her; he/she gave Nisha three more products of price Rs. 1 each. Find out the number of products with Nisha at the end of the transaction.

A 8

B 12

C 11

D 10

Question 81

A pump can be operated both for filling a tank and for emptying it. The capacity of tank is 2400 m^3 .The emptying capacity of the pump is 10 m^3 per min higher than its filling capacity. Consequently, the pump needs 8 min less to empty the tank than to fill it. Find the filling capacity of the pump.

A 45 m^3/min

B 30 m^3/min

C 50 m^3/min

D 55 m^3/min

Question 82

Anand travelled 300 km by train and 200 km by taxi. It took him 5 h and 30 min. However, if he travels 260 km by train and 240 km by taxi, he takes 6 min more. The speed of the train is

A 100km/h

B 120km/h

C 80km/h

D 110km/h

Question 83

Three candidates "A", "B", "C" participated in an election. "A" gets 40% of the votes more than "B". "C" gets 20% votes more than "B". "A" also overtakes "C" by 4000 votes. If 90% voters voted and no invalid or illegal votes were cast, then what will be the number of voters in the voting list?

A 72000

B 80000

C 70000

D 78500

Question 84

Frank, Fardeen, Faulad and Farhan are playing a game where the loser doubles the amount of money that the others have. They manage to play four games. Each player loses a game each as per reverse alphabetical orders of their names. At the end of the game each player was left with ₹32 each. Who started with the most money at the beginning of the game?

A Faulad

B Farhan

C Frank

D Fardeen

Question 85

A function $f(x) = ax^2 + bx + c$, where $a, b, c \in R$, satisfies the property $f(x) < x$ for all $x \in R$. Then which of the following statements must always be **TRUE**?

A $a \leq 0$

B $b \geq 1$

C $c = 0$

D $b > 1$

Question 86

Ajit and Sujit start running on a circular track from the same point in the same direction with speeds a m/s and b m/s. Which of the following cannot be the values of a and b, if the point at which they meet for the fourth time is the starting point?

A 3 m/s and 4 m/s

B 36 m/s and 28 m/s

C 35 m/s and 15 m/s

D 28 m/s and 7 m/s

Question 87

If x is real, the smallest value of the expression $3x^2 - 4x + 7$ is :

A 2/3

B 3/4

C 7/9

D None of the above

Question 88

Two alloys of aluminium have different percentages of aluminium in them. The first one weighs 8 kg and the second one weighs 16 kg. One piece each of equal weight was cut off from both the alloys and first piece was alloyed with the second alloy and the second piece alloyed with the first one. As a result, the percentage of aluminium became the same in the resulting two new alloys. What was the weight of each cut-off piece?

A 3.33 kg

B 4.67 kg

C 5.33 kg

D None of the above

Question 89

Sukriti and Saloni are athletes. Sukriti covers a distance of 1 km in 5 minutes and 50 seconds, while Saloni covers the same distance in 6 minutes and 4 seconds. If both of them start together and run at uniform speed, approximately by what distance will Sukriti win a 5 km mini marathon:

A 50 m

B 200 m

C 100 m

D 300 m

Question 90

A bus operates between cities A and B. The bus stops for a certain time ‘x’ at Canteen. The canteen owner bribes Rs. 12 to the bus driver for every minute he stops at his canteen. While running, the bus travels at a constant speed of 54 kmph. The bus starts from city A at 10 AM and reaches city B at 7 PM. If the amount earned by the driver through bribe is Rs. 1080, then the average speed of journey including stoppage is

A 45 kmph

B 36 kmph

C 40 kmph

D 27 kmph

Question 91

A circular track has 12 markings, increasing in clockwise direction, which are uniformly spaced along the track. 3 people- A, B, and C- started running in the same direction from the same point i.e. point 1. A met B for the first time at point 7 while he met C for the first time at point 4. It is given that B is faster than C. It is also known that B will lose a race against A even when he runs twice as fast as he does now. Where will B meet C for the third time?

A Point 7

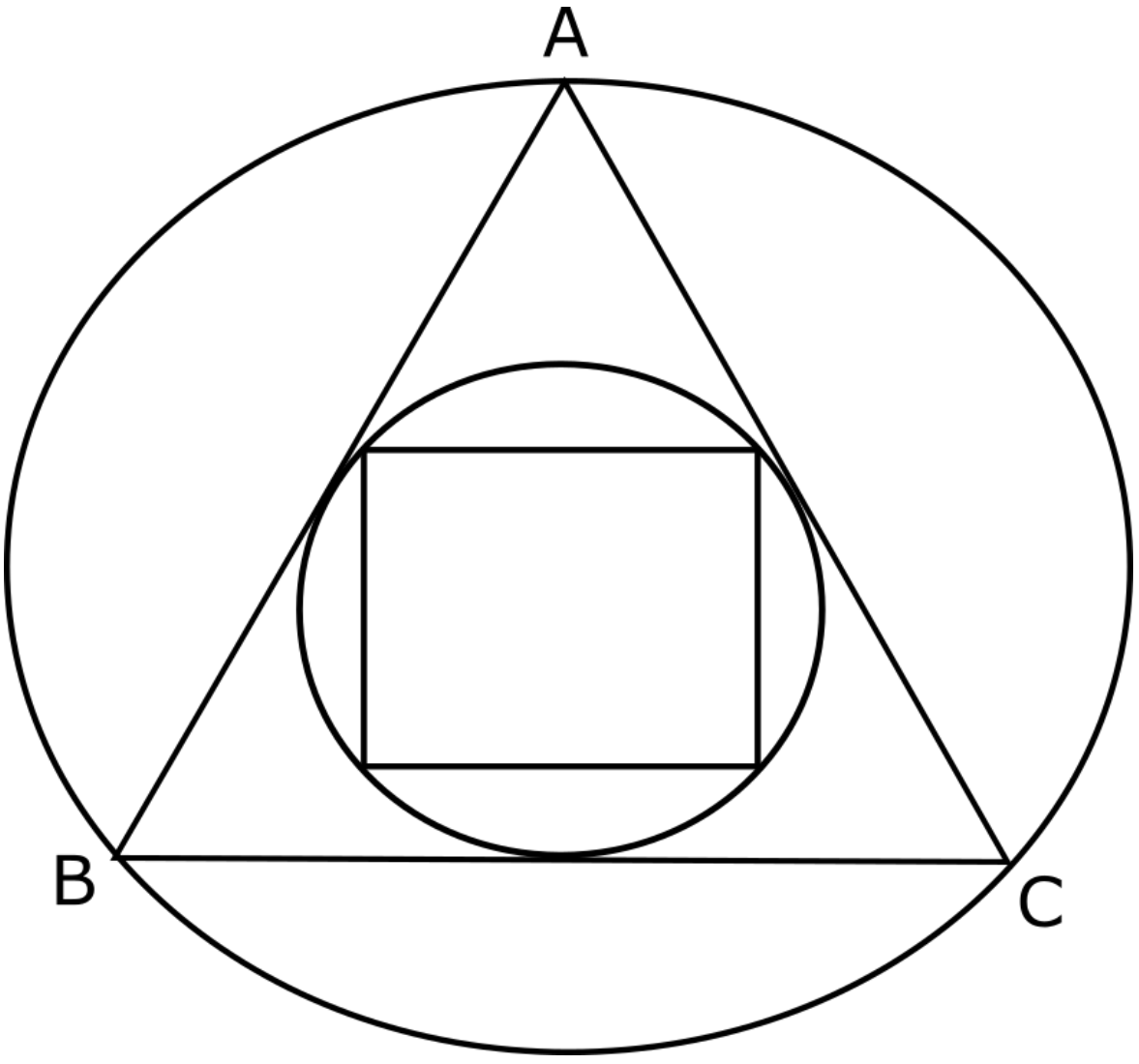
B Between Point 8 and Point 9

C Between Point 4 and Point 5

D Point 1

Question 92

A square is inscribed in the in-circle of an equilateral triangle ABC as shown. What is the ratio of the area of the square and the area of the circumcircle of the triangle ABC?



A $\frac{2}{\pi}$

B $\frac{1}{2\pi}$

C $\frac{1}{4\pi}$

D $\frac{2}{3\pi}$

Question 93

Pipe A of square cross section with side 4cm can fill an empty tank in 30 minutes, Pipe B of a circular cross section with radius 7, is attached to the bottom of the tank, which can empty the tank in 20 minutes. The speeds (in cm/s) at which water flows through the pipes A and B are natural numbers less than 100. Find the volume of the tank.(in m^3) (Assume that $\pi = \frac{22}{7}$)

A 1.76

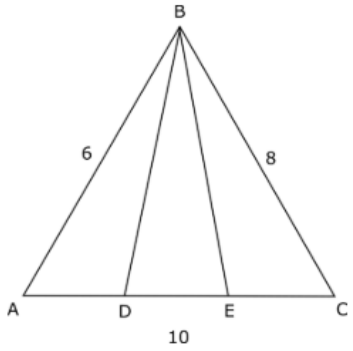
B 2.22

C 2.54

D 3.12

Question 94

In the triangle shown in the figure, AB = 6 cm, BC = 8 cm and AC = 10 cm. AD = DE = EC = 10/3 cm. What is the value of $BD^2 + BE^2$?



A $\frac{500}{9}$

B $\frac{400}{9}$

C $\frac{200}{9}$

D $\frac{100}{9}$

Question 95

A right angled triangle has an inradius 3cm and circumradius 12.5 cm. Find the perimeter of the triangle.

A 54

B 60

C 56

D cannot be determined

Question 96

The ratio of water and milk in three samples is in the ratio 1:3, 2:1 and 3:2. If the three solutions are mixed in the ratio 1:2:3, what is the final ratio of water and milk ?

A 223 : 125

B 5 : 4

C 1 : 2

D 203 : 157

Question 97

The average weight of a class is 54 kg. A student, whose weight is 145 kg, joined the class and the average weight of the class now becomes a prime number less than 72. Find the total number of students in the class now.

A 7

B 13

C 15

D Cannot be determined

Question 98

Two people A and B travel from city X to city Y and from city Y to city X respectively. A leaves city X at 3 PM by car and travels at 70 kmph. B leaves city Y at 4 PM and travels at 60 kmph. If the distance between the cities is 420 km, find the point at which A and B meet.

A 222 km from city Y

B 222 km from city X

C 258 km from city Y

D 258 km from city X

Question 99

Rahul boarded a bus with a 10 rupee note. The bus conductor in a rush to issue tickets, gave Rahul his due but gave rupees instead of paise and paise instead of rupees. Rahul then realized that he had 99 paise less than the amount than he was supposed to have. What was the cost of the bus ticket?

A 3.95

B 7.97

C 5.94

D All of the above

Question 100

The cost of 5 beedis, 7 cigars and 9 cigarettes is 240 Rs. The cost of 8 beedis, 11 cigars and 14 cigarettes is 380 Rs. How much would 1 beedi, 1 cigar and 1 cigarette cost?

A 30

B 36

C 40

D 44

Question 101

Find the in-radius of the triangle with the sides equal to 20 cm, 99 cm, 101 cm respectively.

A 8 cm

B 9 cm

C 11 cm

D 13 cm

Question 102

Ram chases Vijay through a 120 m tunnel. At first, Ram is 80 m before the entrance, and Vijay is somewhere inside. Later, Ram is at the exit, and Vijay is 80 m ahead of him. If the speeds of Ram and Vijay are in the ratio 5 : 3, find Vijay's initial distance from the entrance.

Question 103

What is the maximum value of the product $(15 - p)^9 * (p + 5)^{11}$ where p lies between -5 and 15.

A $11^{10} \times 9^{10}$

B 10^{20}

C $11^{11} \times 9^9$

D $11^9 \times 9^{11}$

Question 104

Given the quadratic equation $x^2 - (A - 3)x - (A - 2)$, for what value of A will the sum of the squares of the roots be zero?

A -2

B 3

C 6

D None of these

Question 105

The ratio of the interior angle to the exterior angle of a regular polygon is 7:2. What is the number of sides of the polygon?

A 12

B 11

C 10

D 9

Answers

1.**D**	2.**2400**	3.**B**	4.**30**	5.**20**	6.**B**	7.**14**	8.**B**
9.**C**	10.**3534**	11.**D**	12.**15**	13.**A**	14.**B**	15.**A**	16.**11**
17.**A**	18.**C**	19.**D**	20.**24**	21.**A**	22.**25**	23.**40**	24.**C**
25.**C**	26.**D**	27.**A**	28.**A**	29.**4**	30.**A**	31.**A**	32.**A**
33.**D**	34.**C**	35.**D**	36.**C**	37.**C**	38.**C**	39.**C**	40.**A**
41.**A**	42.**B**	43.**D**	44.**A**	45.**D**	46.**B**	47.**10**	48.**B**
49.**C**	50.**C**	51.**B**	52.**160**	53.**B**	54.**C**	55.**B**	56.**A**
57.**B**	58.**D**	59.**7**	60.**6**	61.**C**	62.**C**	63.**A**	64.**B**
65.**D**	66.**C**	67.**D**	68.**A**	69.**D**	70.**41**	71.**80**	72.**B**
73.**A**	74.**A**	75.**B**	76.**D**	77.**D**	78.**C**	79.**B**	80.**D**
81.**C**	82.**A**	83.**B**	84.**C**	85.**A**	86.**D**	87.**D**	88.**C**
89.**B**	90.**A**	91.**A**	92.**B**	93.**B**	94.**A**	95.**C**	96.**D**
97.**D**	98.**D**	99.**A**	100.**C**	101.**B**	102.**80**	103.**C**	104.**D**
105.**D**							

Explanations

1.**D**

Let Flash's land speed be x units/sec.

Since Superman takes half the time taken by Flash on the land portion, Superman's speed is twice Flash's land speed:

Superman's speed = $2x$

The land constitutes 40% of the 100-unit track, while the ocean constitutes 60%. Thus,

Land distance=40 and Ocean distance=60

Flash's speed over the ocean is 20% less than his land speed:

$$\text{Flash's ocean speed} = 0.8x$$

Therefore, Flash's total time is

$$= \frac{40}{x} + \frac{60}{0.8x}$$
$$= \frac{40}{x} + \frac{75}{x}$$
$$= \frac{115}{x}$$

Superman travels the entire 100 units at a constant speed of $2x$. Hence, his total time is $= \frac{100}{2x} = \frac{50}{x}$

Given that Flash takes 39 seconds longer:

$$=> \frac{115}{x} - \frac{50}{x} = 39$$
$$=> \frac{65}{x} = 39$$
$$x => \frac{65}{39} = \frac{5}{3}$$

Thus, Flash's land speed= $\frac{5}{3}$, and Superman's speed=$\frac{10}{3}$

Therefore, the required sum is $\frac{5}{3}+\frac{10}{3}=\boxed{5 \text{ units/sec}}$

2. **2400**

Let Raman's usual speed be $10S$ km/hr.

Now, if he had travelled at this speed, he would have covered the distance in 8 hours.

Thus, the total distance $= 80S$ km.

Now, he travels 30% of the distance at 40% of his usual speed.

Time taken $= \frac{24S}{4S} = 6$ hrs

He travels 20% of the distance at 10% of his usual speed.

Time taken $= \frac{16S}{S} = 16$ hrs

He travels the remaining 50% of the distance at 50% of his usual speed.

Time taken $= \frac{40S}{5S} = 8$ hrs

Thus, total time taken to cover the distance $= 6 + 16 + 8 = 30$ hrs.

Thus, he travelled for 30 hrs at an average speed of 80 km/hr.

$\Rightarrow$ Total distance $= 30 \times 80 = 2400$ km.

3. **B**

Let the efficiencies of A,B, and C be a, b, and c respectively.
Objective is to get efficiencies in integers.
Take LCM (12,20) = 60.
Let the total work be 60 units.

$12(a+b) = 60$
$a+b = 5$
$a = 5-b$

$20(b+c) = 60$
$c = 3-b$

The efficiencies of A,B,C are $5-b, b, 3-b$ respectively.

A starts working on the project alone and leaves after 3 days.
Work done by A$= 3(5-b) = 15-3b$
B then takes over and works for 8 days before leaving.
Work done by B$= 8b$
C takes over and finishes the remaining work in exactly 35 days
Work done by C$= 35(3-b) = 105-35b$

Total Work$= 15-3b+8b+105-35b = 60$
$b = 2$

The proportion of work done by A$= \frac{15-3b}{60} = \frac{9}{60}$

The proportion of work done by B$= \frac{8b}{60} = \frac{16}{60}$

The difference in the work of both=$\frac{7}{60}$

Difference in money share= $\frac{7}{60} \times 144000 = 16800$

4. **30**

The question can be solved by backtracking.
The final ratio of milk to water is $4:1$ in 60 liters.
It implies the final milk and water amounts are 48 liters and 12 liters respectively.
This concentration came after 15 liters milk was added. it implies before the 15 liters were drawn off in the new mixture, Milk amount was $48 - 15 = 33$ liters.
It implies after the 15 liters were drawn, Milk and water were 33 liters and 12 liters respectively.

So, 25% is drawn off, it implies 25% of milk and water was drawn off.
Let the milk and water quantity before drawn off be a and b respectively.

$a \times \frac{3}{4} = 33$
$a = 44$ liters

$b \times \frac{3}{4} = 12$
$b = 16$ liters

It implies A and B were mixed to get 44 liter milk and 16 liter water.
Let the quantities of A and B mixed be $5a$ & $3b$ respectively.
A has $4a$ and a as milk and water mixed.
B has $2b$ and b as milk and water mixed.
Equate milk quantity: $4a + 2b = 44$
Equate water quantity: $a + b = 16$

Solve the above to get $a = 6, b = 10$.
It implies amount of A mixed was $5a = 30$ liters.

5. **20**

Let us assume that the efficiencies of a man, a woman, and a child (in units of work per day) are $= x, y$ and z

Let us assume that the total work is w units

Now, a man, a woman, and a child can complete a piece of work together in 20 days.

So, $20(x + y + z) = w$ ...1

Also, a man and a woman can complete the same work in 25 days

$25(x + y) = w$

So, $y = \frac{w}{25} - x$ ...2

And, a woman and a child can complete it in 40 days

$40(y + z) = w$

$z = \frac{w}{40} - y$

$z = \frac{w}{40} - \frac{w}{25} + x$ ...3

Substituting value of y and z from equation 2 and 3 to equation 1

$$20\left(x+\frac{w}{25}-x+\frac{w}{40}-\frac{w}{25}+x\right)=w$$

$$20\left(\frac{w}{40}+x\right)=w$$

$$x=\frac{w}{20}-\frac{w}{40}=\frac{w}{40}$$

Now, let us assume that n men can complete the entire w work in 2 days

So, $2*n*x=w$

$$2*n*\frac{w}{40}=w$$

$$n=20$$

So, 20 men can complete the entire task in 2 days

6. **B**

Let us assume that the length of the initial train given is $=d$ m and its speed is $=v$ m/s

We are given that this train crosses a man running at 5 m/s in the same direction as the train in 15 seconds.

$$\frac{d}{v-5}=15$$

$$d=15(v-5)\ldots1$$

We are also told that the same train crosses another train of length 200 m, moving in the opposite direction at 20 m/s, in 9 seconds.

$$\frac{d+200}{v+20}=9$$

$$d=9(v+20)-200=9v-20\ldots2$$

From 1 and 2, we get:

$$15(v-5)=9v-20$$

$$15v-9v=75-20$$

$$6v=55$$

$v=\frac{55}{6}$ m/s

$$d=9v-20=\frac{9*55}{6}-20=62.5\text{ m}$$

So, the length of the first train is $=62.5$ m

Hence, Option B is correct

7. **14**

Let us assume that the everyday efficiencies of A, B, and C are x, y and z, respectively

We are given that:

$$x=0.5(y+z)$$

$$z=0.5y$$

$$x=0.5*1.5y=0.75y$$

Let us also assume that w is the total work to be done

So, A, B, and C together can complete a piece of work in more than 6 days but less than 15 days

$\therefore \frac{w}{15} < x + y + z < \frac{w}{6}$

$\frac{w}{15} < 0.75y + y + 0.5y < \frac{w}{6}$

$\frac{w}{15} < 2.25y < \frac{w}{6}$

$\frac{4w}{9 * 15} < y < \frac{4w}{6 * 9}$

Dividing the whole equation by w, we get:

$\frac{4}{9 * 15} < \frac{y}{w} < \frac{4}{6 * 9}$ ...1

Now, the number of days taken by B to complete the work alone$= \frac{w}{y}$

So, let's take the reciprocal of equation 1

$\frac{6 * 9}{4} < \frac{w}{y} < \frac{9 * 15}{4}$

$\frac{54}{4} < \frac{w}{y} < \frac{135}{4}$

$13.5 < \frac{w}{y} < 33.75$

So, B will take more than 13 days and less than 34 days

Minimum number of integral days taken by B to complete the task alone$= 14$ days

8. **B**

Let us assume that the cost of 10 litres of pure milk is $=$ Rs $70k$

So, the cost of 1 litres of pure milk is $=$ Rs $7k$

Then, the selling price of 7 litres of milk water solution that he sells would be$=$ Rs $70k$

So, the selling price of 1 litre of milk water solution that he sells would be$=$ Rs $10k$

Now, we are told that the mixture he sells contains 30% water

So, the amount of milk present in 1 litre of milk-water solution is $= 0.7$ litres

So, the actual cost price of 1 litre of milk-water solution is $= 0.7 * 7k =$ Rs $4.9k$

So, the profit percentage of the milk seller$= \frac{10k - 4.9k}{4.9k} * 100 = 104.08\ \%$

Hence, Option B is correct.

9. **C**

Since the two mixtures are mixed in the ratio $a : 5$, we will use a and 5 as the weights for the two solutions to calculate the weighted-average concentration of X.

Concentration of X in the 1st solution $= \frac{11}{18}$.

Concentration of X in the 2nd solution $= \frac{5}{6}$.

$\Rightarrow \frac{\left(\frac{11}{18} \times\ a\right) + \left(\frac{5}{6} \times\ 5\right)}{a + 5} = \frac{a}{a + 1}$

$\Rightarrow (11a + 75)(a + 1) = 18a(a + 5)$

$\Rightarrow 7a^2 + 4a - 75 = 0$

$\Rightarrow 7a^2 + 25a - 21a - 75 = 0$

$\Rightarrow (7a + 25)(a - 3) = 0$

$\Rightarrow a = 3.$

The new mixture has X and Z in the ratio $3 : 1$.

This mixture is mixed with the 2nd initial mixture(5:1) in the ratio $a : a + 1 = 3 : 4$.

The concentration of X will be:

$$\Rightarrow \frac{\left(\frac{3}{4} \times 3\right) + \left(\frac{5}{6} \times 4\right)}{3 + 4} = \frac{67}{84}$$

This concentration in percentage terms is $\frac{67 \times 100}{84} \approx 80\%$.

10. **3534**

Given: $ab + bc = 119$

$\Rightarrow b(a + c) = 7 \times 17$

As both a and c are positive integers, $a + c$ cannot be less than 2.
So, the values that $a + c$ can take are $119, 17$ or 7.

Case 1: $a + c = 119$
$\Rightarrow b = 1$
As $a - c = 5$, we get $a = 62$ and $b = 57$. The value of $abc = 3534$.

Case 2: $a + c = 17$
$\Rightarrow b = 7$
As $a - c = 5$, we get $a = 11$ and $b = 6$. The value of $abc = 462$.

Case 3: $a + c = 7$
$\Rightarrow b = 17$
As $a - c = 5$, we get $a = 6$ and $b = 1$. The value of $abc = 102$.

The maximum possible value of abc is 3534.

11. **D**

Let us assume that the total amount of pure milk is $8v$ litres

So, the amount of pure milk in container A initially is $= 3v$ litres

Now, the amount of milk left in container A after replacement is $= 3v(1 - 0.1)^3 = 3 * 0.729v = 2.187v$

Now, the amount of pure milk in container A initially is $= 5v$ litres

Now, the amount of milk left in container B after replacement is $= 5v(1 - 0.3)^3 = 5 * 0.343v = 1.715v$

So, now let us assume that the cost price of 1 litre of pure milk for the milk seller is $=$ Rs 1

So, the profit percentage of the milk seller will be $= \frac{(3v + 5v) - (2.187v + 1.715v)}{2.187v + 1.715v} * 100 \approx 105.02\%$

Hence, Option D is correct

12. **15**

We are given that a and b are non-negative integers, i.e. $a, b \geq 0$

We are given:

$\Rightarrow \frac{2}{a} + \frac{3}{4b} = \frac{1}{4}$

$\Rightarrow 8b + 3a = ab$

$\Rightarrow 8b + 3a - ab = 0$

$\Rightarrow 8b - ab + 3a - 24 = -24$

$\Rightarrow (8 - a)(b - 3) = -24$

$\Rightarrow (a - 8)(b - 3) = 24$

Now, we need to express 24 as a product of two numbers.
Let us assume that $a - 8 = m$, and $b - 3 = n$, such that $mn = 24$.

Sum of a and b is $m + n + 11$. So, to find the maximum sum, we need to maximise $m + n$. This is possible only when we take them as 24 and 1.
So, the maximum value of $(a + b)$ is 36.

The minimum value occurs when m and n are as close as possible.

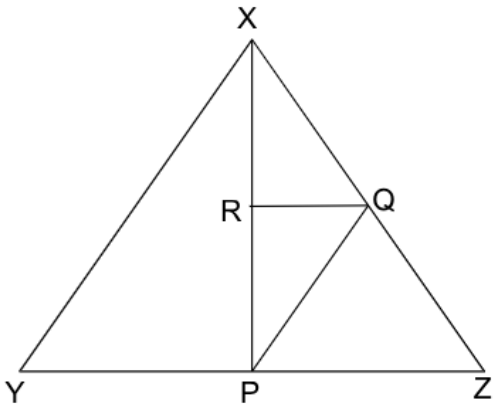
That will be m = 6 and n = 4

So, the minimum value of $(a + b)$ is 21.

And the absolute difference between the maximum and minimum value is $= 36 - 21 = 15$.

13. **A**

The figure for this triangle will look like:



In tirangle XPQ, QR is the median. Using Apollonius theorem, $XQ^2 + QP^2 = 2\left(QR^2 + RP^2\right)$.

We know that PQ is the median on XZ, so XQ $= \frac{1}{2} \times XZ = 9$ cm.

P and Q are both mid points, so PQ will be half of XY.(Using mid-point theorem)
So, PQ $= 8$cm.

We need RP which is half of XP.

Now, XP is a median, so we can again use Apollonius theorem in triangle XYZ:

$\Rightarrow XY^2 + XZ^2 = 2\left(XP^2 + PZ^2\right)$

PZ will be half of YZ, i.e. 10 cm.

$\Rightarrow 16^2 + 18^2 = 2\left(XP^2 + 10^2\right)$

$\Rightarrow$ XP $= \sqrt{190}$

From this, RP $= \frac{\sqrt{190}}{2}$

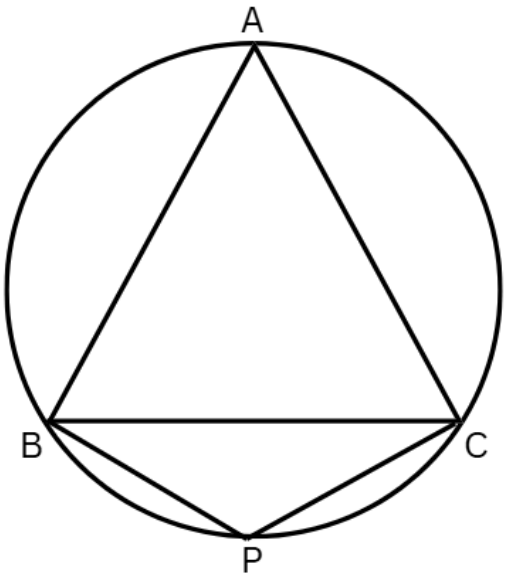
We put all these values in the 1st equation to calculate the value of QR.

$\Rightarrow 8^2 + 9^2 = 2\left[QR^2 + \left(\frac{\sqrt{190}}{2}\right)^2\right]$

$\Rightarrow$ QR $= 5$ cm.

Hence, Option A is correct.

14. **B**



ABPC is a cyclic quadrilateral.

Let the side of the triangle be s cm each

Apply Ptolemy's theorem,

$AP * BC = (AB * PC) + (AC * BP)$

$AP * s = s * 5 + s * 3$

$AP = 8$ cm.

$\angle BPC = 180^\circ - \angle BAC = 120^\circ$

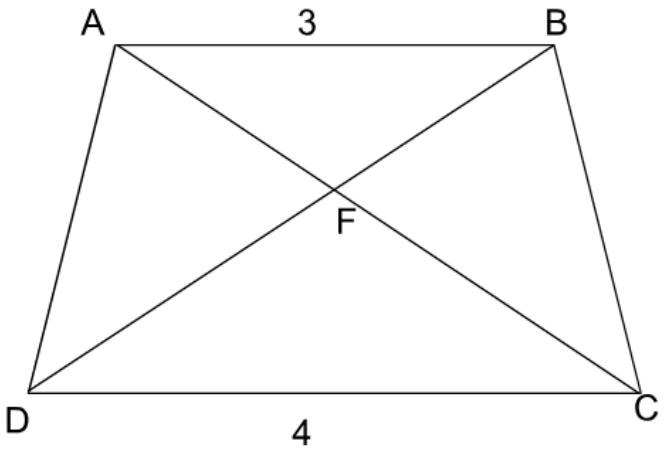
Apply cosine rule in BPC,

$BC^2 = 3^2 + 5^2 - 2 \times 3 \times 5 \times \cos 120^\circ = 49$

$BC = 7$ cm.

$AP^2 + BC^2 = 64 + 49 = 113$

15. **A**



Consider $\triangle AFB, \triangle DFC$

$\angle BAF = \angle FCD$(Interior alternate angles)

$\angle ABF = \angle FDC$(Interior alternate angles)

$\angle AFB = \angle DFC$(Vertically opposite angles)

We can see that the triangles are similar to each other.

$$\frac{\mathrm{AB}^2}{\mathrm{CD}^2} = \frac{\text{Area of AFB}}{\text{Area of DFC}}$$

$$\frac{3^2}{4^2} = \frac{9}{\text{Area of DFC}}$$

Area of DFC = 16

Let the perpendicular distance of F from AB be h_1, and the perpendicular distance of F from CD be h_2.

Since the area of triangle AFB is 9 sq. cm,

$$\text{Area of AFB} = \frac{1}{2} \times AB \times h_1$$

$$9 = \frac{1}{2} \times 3 \times h_1$$

$$h_1 = 6 \text{ cm}$$

Similarly, the area of triangle DFC is 16 sq. cm.

$$\text{Area of DFC} = \frac{1}{2} \times CD \times h_2$$

$$16 = \frac{1}{2} \times 4 \times h_2$$

$$h_2 = 8 \text{ cm}$$

Since AB $||$ CD, the perpendicular distances h_1 and h_2 together make up the height of the trapezium.

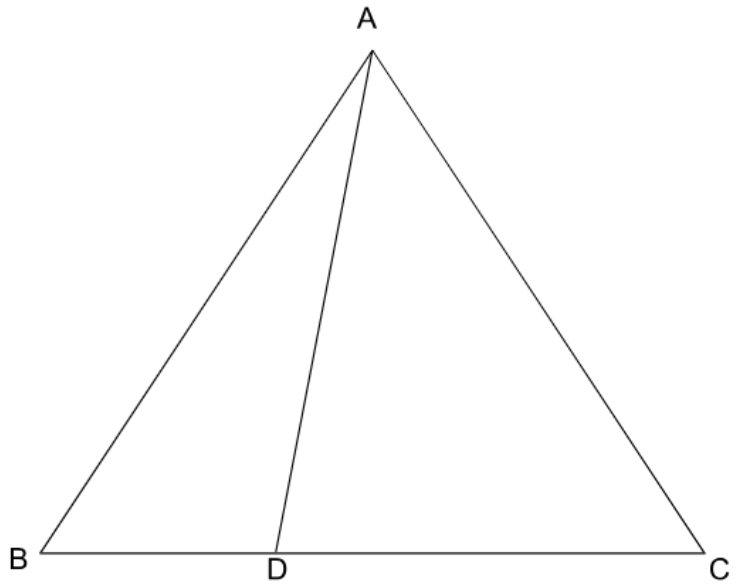
$$\text{Height of trapezium} = h_1 + h_2$$

$$= 6 + 8$$

$= \boxed{14 \text{ cm}}$

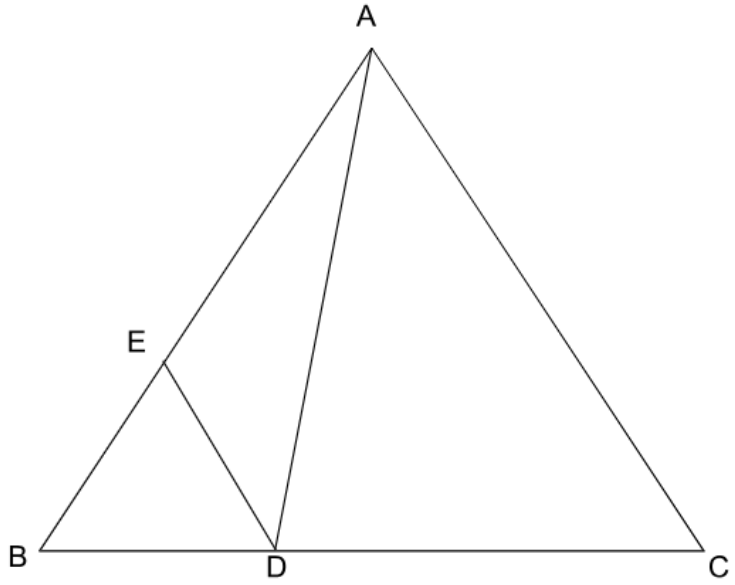
Therefore, the height of the trapezium is 14 cm.

16. **11**



AD divides BD and DC in the ratio 2:3.
This means the areas of triangles ABD and ADC are also in the ratio 2:3.

So, the areas of triangles ADB and ADC will be 10 and 15 sq cm, respectively.

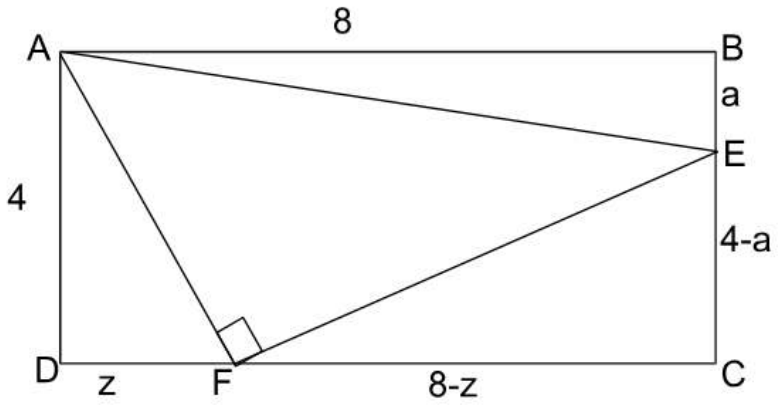


Now, the base AB is divided in the ratio 2:3, which means that the ratio of the areas of triangles BED and AED will be 2:3.

So, the area of triangle DEB = $\frac{2}{5} \times 10 = 4$ sq. cm.

The difference in the areas of triangles DEB and ADC $= 15 - 4 = 11$ sq. cm.

17. **A**



Consider $\triangle ADF$. Let us assume $\angle DAF = \theta$

$\angle AFD = 90 - \theta$

$\angle EFC = \theta$ (Angles on a straight line)

$\angle CEF = 90 - \theta$

Using AAA criterion for similarity we can see that

$\triangle ADF \sim \triangle FCE$

Now, we are also given that ratio of the areas of $\triangle ADF, \triangle FCE = 4 : 9$

In similar triangles we know that, ratio of the areas is the square of the ratio of their sides

$$\implies \frac{\mathrm{AD}^2}{\mathrm{FC}^2} = \frac{\mathrm{DF}^2}{\mathrm{CE}^2} = \frac{4}{9}$$

$$\implies \frac{4}{8-z} = \frac{z}{4-a} = \frac{2}{3}$$

Solving we get

$z = 2, a = 1$

Using pythagoras theorem, we can find

$AF = \sqrt{4^2 + 2^2} = \sqrt{20}$

$FE = \sqrt{6^2 + 3^2} = \sqrt{45}$

Thus, area of $\triangle AFE = \frac{1}{2} \times AF \times FE = 15\mathrm{cm}^2$

18. **C**

In the race between P and Q, Q ran only 550 metres, which took him $\frac{550}{25} = 22$ seconds. This means that P took $22 - 10 = 12$ seconds to finish the race of 600 metres, which gives his speed $= \frac{600}{12} = 50$ m/sec.

In the same race, Q won against R by 6 seconds, which means that for the entire race R took $= \frac{600}{25} + 6 = 30$ seconds and the speed of Q will be $\frac{600}{30} = 20$ m/sec.

Now, P and R are racing in a 1000-meter race.

P will take $\frac{1000}{50} = 20$ seconds, and R will take $\frac{1000}{20} = 50$ seconds.

So, R will lose by $50 - 20 = 30$seconds.

19. **D**

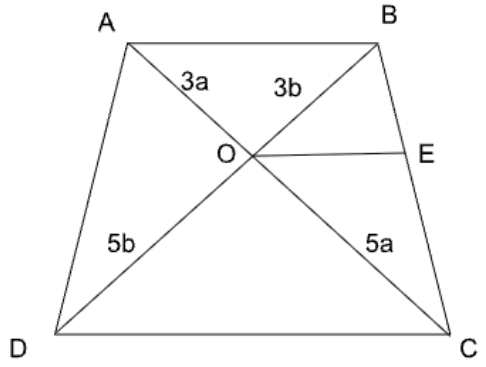
Let us assume that $AB = 3m$ cm then $CD = 5m$ units

Now, since AB and CD are parallel sides, this makes triangles AOB and COD similar triangles

So, $\frac{AO}{OC} = \frac{BO}{OD} = \frac{AB}{CD} = \frac{3}{5}$

So, let us assume that $AO = 3a$ then $OC = 5a$

And let us also assume that $OB = 3b$ then $OD = 5b$



Now, $\frac{\text{Area of triangle AOB}}{\text{Area of triangle COD}} = \frac{3^2}{5^2} = \frac{9}{25}$

Let us assume that the area of triangle $AOB = 9X$ then area of triangle $COD = 25X$

Now, the triangles BOC and COD share the same height

So, if Area of triangle COD is $25X$ then the area of triangle BOC will be $= \frac{3 * 25X}{5} = 15X$

Now, since OE is parallel to DC so by Basic proportionality theorem

$\frac{OB}{OD} = \frac{BE}{EC} = \frac{3}{5}$

Now, triangles BOE and COE share the same height, so their areas are in the same ratio as their base lengths BE and EC

$\frac{\text{Area of triangle BOE}}{\text{area of triangle COE}} = \frac{BE}{EC} = \frac{3}{5}$

So, the area of triangle COE will be $= \frac{5 * \text{Area of triangle BOE}}{3}$

Also, the sum of areas of triangle BOE and COE is equal to area of the complete triangle BOC which was $15X$

$\frac{8 * \text{Area of triangle BOE}}{3} = 15X$

$\therefore \text{Area of triangle BOE} = \frac{45X}{8}$

Hence, $\text{Area of triangle BOE: Area of triangle BOC} = \frac{45X}{8} : 15X = 3 : 8$

Hence, option D is correct.

20. **24**

Let the total work be 60 units.
Let the efficiency of A,B, and C be a, b, and c respectively.
$a + b = 5$ -----> Eq(1)
$b + c = 4$ -----> Eq(2)

A works alone for 4 days, after which B joins A. After working together for 3 days, A's efficiency increases by 25% because of a process improvement. A and B then work together for another 4 days, after which B leaves and C joins A. A's efficiency now decreases by 20%. A and C together complete the remaining work in 2 days.
A works at efficiency a for first 7 days, $1.25a$ for next 4 days and at $1.25a * 0.8 = a$ for next 2 days.

$4a + 3(a + b) + 4(1.25a + b) + 2(a + c) = 60$
$14a + 7b + 2c = 60$ ----->Eq(3)

Solve Eq(1),(2) & (3):
$a = 3, b = 2, c = 2.$

If A and C work together alternatively, they will complete 5 units in every 2 day period.
It implies they will complete the job in $12 * 2 = 24$ days

21. **A**
The average of the four numbers is 37, which implies that:
$$\frac{a + b + c + d}{4} = 37$$
$a + b + c + d = 148$---- 1st eq

After the increase, the average becomes 44.15, or the sum becomes $4 \times 44.15 = 176.6.$

$1.4a + 1.3b + 1.2c + 1.1d = 176.6.$

We subtract $1.1 \times 1st$ eq from this to get

$0.3a + 0.2b + 0.1c = 13.8$

$3a + 2b + c = 138.$

Now, we want to maximise a^3b^2c, which is nothing but:
$a \cdot a \cdot a \times \ b \cdot b \times \ c$

Using the AM-GM inequality:
$$\frac{a + a + a + b + b + c}{6} \geq \sqrt[6]{a^3b^2c}$$
The LHS is nothing but $\frac{3a + 2b + c}{6} = \frac{138}{6} = 23.$
So, the GM or $\sqrt[6]{a^3b^2c}$, will be maximum when all the terms are equal, i.e.,

$a = a = a = b = b = c = 23.$

To find the value of d, we simply put these values into the 1st equation, which gives $d = 79.$

Hence, the correct answer is Option A.

22. **25**
Let us assume that the walking speed of a person is $= 4a$ steps per sec

Then his running speed will be $= 9a$ steps per sec

Now, let us assume that there are n steps on the escalator and the speed of the escalator is e steps per sec

We were told that the person walks up an upward-moving escalator and reaches the top in 15 seconds

$$\frac{n}{e+4a} = 15$$

or, $n = 15e + 60a$ ...1

We were also given that if he runs up the same escalator, he reaches the top in 10 seconds.

$$\frac{n}{e+9a} = 10$$

or, $n = 10e + 90a$ ...2

From equations 1 and 2, we get:

$15e + 60a = 10e + 90a$

$5e = 30a$

or, $e = 6a$

or, $a = \frac{e}{6}$

Now, substituting the value of a in first equation, we have:

$$n = 15e + \frac{60 * e}{6}$$

$n = 25e$

or, $\frac{n}{e} = 25$ sec

So, if the person is stationary, then he will reach the top in 25 sec

23. **40**

Let us assume Manoj and Rushi cover 3x and 5x metres per hop.

The length of the race is equal to 60 hops of Rushi, i.e., $60 \times$ 5x = 300x.

Now, assume Manoj hopped N times before Rushi started, and the race ended in a dead heat.

This implies that in the time Rushi covered the full 300x distance, Manoj covered only 300x - N*3x.

As they both took an equal amount of time to cover these distances, the ratio of distance covered will be equal to the ratio of their speeds.

$$\frac{300x}{300x - 3Nx} = \frac{5}{3}$$

$$900x = 1500x - 15Nx$$

$$15Nx = 600x$$

N = 40 hops.

So, the correct answer is 40.

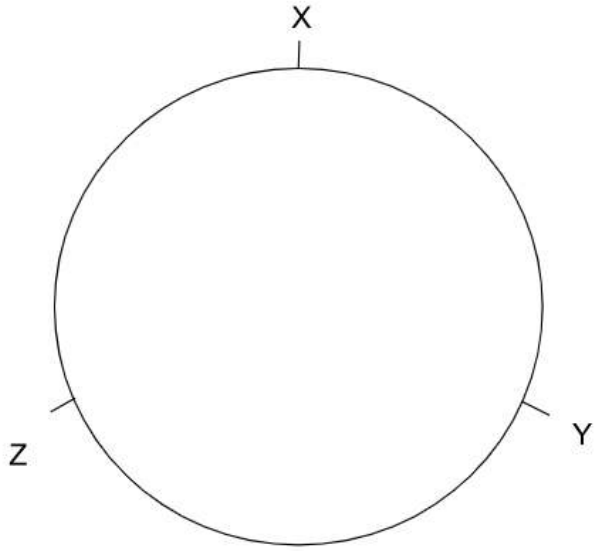
24. **C**

Amal and Bimal are running in the same direction with their speeds in the ratio 4:7. So, they will meet at |4 - 7|, i.e., 3 distinct points.

Chetan is running in the opposite direction of Amal but twice his speed, which means that the ratio of their speeds is 2:1. This implies that they will meet at (2 + 1) = 3 distinct points.

Now, as Chetan runs at twice the speed of Amal and the ratio of speeds of Amal and Bimal is 4:7, the ratio of speeds of Chetan and Bimal is 8:7. This implies that they will meet at (8 + 7) = 15 distinct points.

As all the meeting points are equidistant, we can divide the track into 15 and 3 equal areas:



X, Y and Z are the 3 points on the track that will be common in all the pairwise meeting points.
Since they are counted thrice (in 3 pairs), we must subtract them twice, giving a total of (3 + 3 + 15 - 2*3) = 15 distinct meeting points.

25. **C**

Let us assume that the speed of A is $= 2k$ m/s

Then speed of B and C will be $3k$ m/s and $5k$ m/s

Now, the time taken by C to finish the race$= \frac{100}{5k} = \frac{20}{k}$

And the distance travelled by A when C finishes the race with a 10 m headstart will be

$= 2k * \left(\frac{20}{k}\right) + 10 = 40 + 10 = 50$ m

Now, since B had a 5 sec head start, so distance travelled by B when C finishes is

$= 3k * \left(\frac{20}{k} + 5\right) = 60 + 15k$ m

Now, since C is the first person to complete the race, $60 + 15k$ m should be less than 100 m

Hence, the only acceptable values of k are 1 and 2

Now, we will have to proceed case-wise from here:

CASE 1: $k = 1$

Then the speed of A, B and C is 2 m/s , 3 m/s and 5 m/s ,respectively

So, B will complete the race in $\frac{100}{3}$ sec [inclusive of headstart advantage]

And, when B finishes, A will have covered a distance $= 2\left(\frac{100}{3} - 5\right) + 10 = \frac{200}{3}$ m

Hence, B defeats A by $\frac{100}{3}$ m

CASE 2: $k = 2$

Then the speed of A, B and C is 4 m/s , 6 m/s and 10 m/s ,respectively

So, B will complete the race in $\frac{100}{6}$ sec [inclusive of headstart advantage]

And, when B finishes, A will have covered a distance $= 4\left(\frac{100}{6}-5\right)+10 = \frac{170}{3}$m

Hence, B defeats A by $\frac{130}{3}$ m

Hence the maximum possible distance by which b defeats A is $\frac{130}{3}$ m

Therefore, Option C is correct.

26. **D**

The final concentration of milk is given as $33.33\% = \frac{100}{3}$.

Since the initial volume of the container is 72 L, the volume of milk present in the final solution is $\frac{100\times 72}{3\times 100} = 24$L.

After consecutive replacements of $24, x$ L we have 24 L left finally. Thus we can write

$$72\left(1-\frac{24}{72}\right)\left(1-\frac{x}{72}\right)=24$$
$$\frac{48}{72}\left(1-\frac{x}{72}\right)=\frac{24}{72}$$
$$\left(1-\frac{x}{72}\right)=\frac{1}{2} \implies x = 36 \text{ L}$$

27. **A**

V_r is the volume replaced every time

V_t is the total volume of solution

n is the number of replacements

Then

Remaining quantity of substance = Initial Volume $\left(1-\frac{V_r}{V_t}\right)^n$

$$27 = 64\left(1-\frac{16}{64}\right)^n$$
$$\frac{27}{64} = \left(\frac{3}{4}\right)^n$$

Thus, $n = 3$

28. **A**

Since $20 < 25 < 40$ and $20 < 25 < 36$, we consider the alligations as follows

$$\frac{Q_A}{Q_C} = \frac{40-25}{25-20} = \frac{3}{1}$$
$$\frac{Q_A}{Q_B} = \frac{36-25}{25-20} = \frac{11}{5}$$

Adding the portions of solution A in both we end up with the final ratios of the solutions as

$14:5:1$

29. **4**

We are given the equation: $(2-\sqrt{3})^{x^2-2x-2}+(2+\sqrt{3})^{x^2-2x-2}=4$

Let us assume that $a=(2-\sqrt{3})^{x^2-2x-2}$

Then, $\frac{1}{a}=\left(\frac{1}{2-\sqrt{3}}\right)^{x^2-2x-2}=(2+\sqrt{3})^{x^2-2x-2}$

Now, we can rewrite the given equation as:

$a+\frac{1}{a}=4$

or, $a^2-4a+1=0$

or, $a=\frac{4\pm\sqrt{16-4}}{2}$

or, $a=2\pm\sqrt{3}$

Now, we get 2 cases from here:

CASE 1: $a=2-\sqrt{3}$

or, $(2-\sqrt{3})^{x^2-2x-2}=(2-\sqrt{3})^1$

or, $x^2-2x-2=1$

or, $x^2-2x-3=0$

From here, we get: $x=-1$ or $x=3$

CASE 2: $a=2+\sqrt{3}$

or, $(2-\sqrt{3})^{x^2-2x-2}=(2+\sqrt{3})^1=(2-\sqrt{3})^{-1}$

or, $x^2-2x-2=-1$

or, $x^2-2x-1=0$

From here, we get: $x=1+\sqrt{2}$ or $x=1-\sqrt{2}$

Hence, the sum of all possible real values of x is $=-1+3+1+\sqrt{2}+1-\sqrt{2}=4$

30. **A**

Let the speed of the faster car be v kmph and the distance between point A and B be x km. The time taken in hours is

$1+\frac{2}{60}+\frac{30}{3600}=\frac{25}{24}$ hours

$\frac{25}{24}=\frac{x}{v}\implies v=\frac{24}{25}x$

Distance travelled by the slower car during this time is

$\frac{25}{24}\times 24=25$km

Now the distance between the two cars is $(x-25)$km. Since they are facing opposite directions, using relative speed, we can write

$\text{time to meet}=\frac{x-25}{24+v}$

$\frac{20}{60}+\frac{50}{3600}=\frac{25}{72}=\frac{x-25}{24+v}$

$\frac{25}{72}(24+\frac{24}{25}x)=x-25$

$$\frac{25}{3}(1+\frac{1}{25}x) = x - 25$$
$$25 + x = 3(x - 25)$$
$$x = 50 \text{ km}$$

31. **A**

Given initially

$$M + O + G = 145$$

After selling the fruits

$$\frac{4}{5}M + \frac{1}{2}O + \frac{1}{3}G = 66$$

Since we can only sell a natural number of fruits, G must be a multiple of 3 as well, and O must be a multiple of 2, and M must be a multiple of 5.

Thus, G must be a multiple of 75 and O must be a multiple of 50 (given that they are multiples of 25 in the question)

Hence, G can be $75, 150, ..$ while O can be $50, 100, ..$

From the original equation, we can see that $O + G$ must be less than 145. Thus the only values of O and G that satisfy this are $(50, 75)$

Thus, $M = 145 - (75 + 50) = 20$

32. **A**

$$\frac{(x^2+1)^2 + 9x^2}{x(x^2+1)} = \frac{x^2+1}{x} + \frac{9x}{x^2+1}$$

Let's assume $\frac{x^2+1}{x} = t$

Now the equation reduces to the form

$$t + \frac{9}{t}$$

To find the minima, we can use the AM-GM inequality. We know that the lowest value AM takes is when the terms are equal.

Thus,

$t = \frac{9}{t}$ or $t = \pm 3$

We know that $x > 1$ hence t cannot be negative and we must solve for

$$\frac{x^2+1}{x} = 3$$
$$x^2 - 3x + 1 = 0$$

Solving using the quadratic formula, we end up with two values of x

$$x = \frac{3 \pm \sqrt{5}}{2}$$

We reject $x = \frac{3-\sqrt{5}}{2}$ as it's mentioned in the question that x>1.

Thus, $x = \dfrac{3+\sqrt{5}}{2}$

33. **D**

Using the Sine rule we have:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

where : a, b, c are triangle side lengths; A, B, C are the measures of their respective opposite angles ; R is the circumradius.

Substituting the given values for one side, we get:

$$\frac{4}{\sin(30^\circ)} = 2R$$

Solving for R, we get $R = 4$ cm

34. **C**

As CE bisects $\angle BCD$, which is the exterior angle of triangle ABC, we can use the exterior angle bisector theorem, to show that:

$$\frac{AE}{BE} = \frac{AC}{BC}$$

We can observe that $AE = AB + BE$, so substituting it in above equation we get:

$$\frac{AE}{BE} = \frac{AB+BE}{BE} = \frac{AB}{BE} + 1$$

It is given that $\dfrac{AB}{BE} = \dfrac{2}{3}$,

Thus, $\dfrac{AE}{BE} = \dfrac{2}{3} + 1 = \dfrac{5}{3}$

We already know that, $\dfrac{AE}{BE} = \dfrac{AC}{BC}$ and $AC = 5$ cm. Thus, we get:

$$\frac{5}{3} = \frac{5}{BC}$$

Thus using the theorem we get $BC = 3$ cm.

35. **D**

A polygon of n sides has $\dfrac{n(n-3)}{2}$ diagonals.

$$\frac{n(n-3)}{2} = 350$$

$$n^2 - 3n - 700 = 0$$

$$n^2 - 28n + 25n - 700 = 0$$

$$(n-28)(n+25) = 0$$

Using the quadratic formula, we get two values of n: $28, -25$. We reject -25 as a polygon cannot have negative number of sides. Hence, 28 is the answer.

36. **C**

We are given that the triangles are isosceles triangles with all integral side lengths, with 20 cm as their perimeter.

Let us assume the two equal sides are of length a cm and the third unequal side is of length b cm.

Since, Perimeter of Triangle$= a + a + b = 20$

or, $b = 20 - 2a$

Now as b is an integer and length of a side, it must be positive.

So, we have $b > 0$ which means $20 - 2a > 0$

Thus, $10 > a$

So the largest possible value of a can be 9 cm and as it is an integer and length of a side, it's minimum value is 1cm.

Now let's try to make cases:

CASE 1: $a = 9$ cm

Then, $b = 2$

Now, let's check if such a triangle follows the triangle inequality

So, $|9 - 9| < (b = 2) < |9 + 9|$

Thus, triangle with $(9, 9, 2)$ side lengths is one possible case

Case 2: $a = 8$ cm

Then, $b = 4$

Now, let's check if such a triangle follows the triangle inequality

So, $|8 - 8| < (b = 4) < |8 + 8|$

So, triangle with $(8, 8, 4)$ side lengths is another possible case

Case 3: $a = 7$ cm

Then, $b = 6$

Now, let's check if such a triangle follows the triangle inequality

So, $|7 - 7| < (b = 6) < |7 + 7|$

So, triangle with $(7, 7, 6)$ side lengths is another possible case

Case 4: $a = 6$ cm

Then, $b = 8$

Now, let's check if such a triangle follows the triangle inequality

So, $|6 - 6| < (b = 8) < |6 + 6|$

So, triangle with $(6, 6, 8)$ side lengths is another possible case

Case 5: $a = 5$ cm

Then, $b = 10$

Now, let's check if such a triangle follows the triangle inequality

So, $|5 - 5| < (b = 10) = |5 + 5|$

So, a triangle with $(5, 5, 10)$ side lengths is not a valid case, as this case does not follow the triangle inequality

We can observe from above case that as b continues to increase further while a decreases, the triangle inequality will not be followed. Thus, all the remaining cases will violate the triangle inequality and can be ruled out.

So, there are only 4 possible triangles: $(9, 9, 2), (8, 8, 4), (7, 7, 6), (6, 6, 8)$.

Alternative Solution:

Since p = 20 is even:

Total number of triangles = $\left[\frac{p^2}{48}\right]$ = [400/48] = [8.33] → **8**

Number of scalene triangles = $\left[\frac{(p-6)^2}{48}\right]$ = [196/48] = 4.08 → **4**

Non-scalene triangles = Total − Scalene = 8 − 4 = 4

A non-scalene triangle is either isosceles or equilateral. An equilateral triangle would need side length 20/3, which is not an integer, so none exist.

Hence, the number of isosceles triangles = **4**.

Therefore, Option C is correct.

37. **C**

We are given the ratios.

Let us assume amount mixed of alloy1, alloy 2, and alloy 3 be $10x$, $10y$, and $10z$ respectively by the alloy maker.

So, the amount mixed of Copper, Zinc, and Tin from Alloy 1 is $2x$, $3x$, and $5x$ respectively.

The amount mixed of Copper, Zinc, and Tin from Alloy 2 is $4y$, y, and $5y$ respectively.

The amount mixed of Copper, Zinc, and Tin from Alloy 3 is $3z$, $4z$, and $3z$ respectively.

We are given that an alloy maker mixes alloy 1, alloy 2, and alloy 3 in the ratio 4:5:6.

Let the amount mixed of alloy 1, alloy 2, and alloy 3 be $4a$, $5a$, and $6a$ respectively.

Now, $10x = 4a, 10y = 5a, 10z = 6a$

So, $x = \frac{2a}{5}, y = \frac{a}{2}, z = \frac{3a}{5}$

Given the Tin present in mixture is 252 gram

Total Tin present will be sum of Tin present in Alloy 1, Alloy 2, and Alloy 3.

Total Tin$= 5x + 5y + 3z = 2a + \frac{5a}{2} + \frac{9a}{5} = \frac{63a}{10} = 252$

So,$a = 40$ gram.

Amount of Copper present$= 2x + 4y + 3z = \frac{4a}{5} + 2a + \frac{9a}{5} = \frac{23a}{5} = 184$ gram.

Option C.

38. **C**

Let the batsman's scores from match 1 to match 8 be a, b, c, d, e, f, g, h in that order.

$a < b < c < d < e < f < g < h$

We need to minimise e but $f, g, h \geq\ e\ \geq\ a, b, c, d$. It means that we need to minimse a, b, c, d and maximise f, g, h.

We are given the average of his scores from match 1 to match 8 is 21.5.

$a + b + c + d + e + f + g + h = 8 * 21.5 = 172$ ----> Eqn(2)

The average of his scores in the first six matches, and in the last five matches is 19.833, and 23.6 respectively.

$a+b+c+d+e+f=6*19.833=119$ -----> Eqn(3)
$d+e+f+g+h=5*23.6=118$ -----> Eqn(4)

Subtracting Eqn(3) from Eqn(2), we get
$g+h=53$ -----> Eqn(5)
We know that $g<h$.
Greatest value of g occurs when the difference between g and h is 1.
So, $g \leq 26,\ h \geq 27$ ---->Eqn(6)

Subtracting Eqn(4) from Eqn(2),
$a+b+c=54$ ----> Eqn(7)
Least value of c occurs when a, b, c are close to each other. $17, 18, 19$ is one such possibility.
So, $c \geq 19$ -----> Eqn(8)

Subtracting Eqn(5) from Eqn(4),
$d+e+f=65$ -----> Eqn(9)
We know $f<g$ and $c<d$.
So, $f \leq 25$ and $d \geq 20$
$d<e$.
So, the least value of $e=21$.

Hence, Option C is the correct answer.
39. **C**
We are given that the pair of linear equations $ax+6y=13$ and $7x+by=7$ has no solution.

We know that this pair of linear equation has no solutions if:

$\frac{a}{7}=\frac{6}{b} \neq \frac{13}{7}$

So $\frac{a}{7}=\frac{6}{b}$

or $ab=7*6$

or $ab=2*3*7$

Therefore, the number of factors of the product ab is$=(1+1)(1+1)(1+1)=8$

Therefore, $k=8$

Hence, Option C is correct.

40. **A**
We are given that the train of length 1050 m is moving at a speed of 10 m/s. A car starts from the back end (caboose) of the train and travels along the train at a speed of 25 m/s (with respect to the ground). It reaches the front end (engine) of the train, immediately turns around, and travels back to the caboose.

Let us assume that the train is at rest.

Then, for the first part of the journey, when the car travels from the caboose to the engine of the train, its relative speed with respect to the train will be $=25-10=15$ m/s

Now, in this first part of the journey, the car travels the entire length of the train from the caboose to the engine.

Hence, the time taken by the car to cover this first part of the journey is $=\frac{1050}{15}=70$ sec.

Now, when the car reaches the engine, it reverses its direction and travels from the engine to the caboose.

Then, for the second part of the journey, when the car travels from the engine to the caboose of the train, its relative speed with respect to the train will be $=25+10=35$ m/s

Now, in this journey too, the car travels the entire length of the train from the engine to the caboose.

Hence, the time taken by the car to cover this second part of the journey is $= \frac{1050}{35} = 30$ sec.

Now, the total time taken by the cal to complete the entire journey$= 70 + 30 = 100$ sec

Therefore, in any relative reference frame, the car completes the entire journey in 100 sec.

Now, in our initial frame of reference, we were given that speed of the train$= 10$ m/sec

Therefore, the train distance travelled by train in 100 sec$= 10 * 100 = 1000$ m

Hence, Option A is correct.

41. **A**

Let us assume that the person's usual speed is v km/hr and the usual time he takes to cover the distance is t hours.

Therefore, the distance travelled by the person will be $= vt$ km

Now, we are given that if he were to travel at 6 km/h slower than his usual speed, he would take 4 hours longer to complete the same distance.

Therefore, the distance travelled by the person can be written as $= (v - 6)(t + 4)$ km

Also, we are given that if he were to travel at 4 km/h faster than his usual speed, he would take 1 hour less than the usual time.

Therefore, the distance travelled by the person can be written as $= (v + 4)(t - 1)$ km

We know that the distance travelled by the person in each case is the same.

$\therefore vt = (v - 6)(t + 4) = (v + 4)(t - 1)$

Now firstly, we have

$vt = (v - 6)(t + 4)$

or $vt = vt + 4v - 6t - 24$

or $4v = 6t + 24$

or $2v = 3t + 12$....(1)

Secondly, we have,

$vt = (v + 4)(t - 1)$

$vt = vt - v + 4t - 4$

$v = 4t - 4$.....(2)

From 1 and 2, we get

$3t + 12 = 2(4t - 4)$

$3t + 12 = 8t - 8$

$5t = 20$

$t = 4$ hours

Also, $v = 4t - 4 = 4(4) - 4 = 12$ km/hr

Hence, the usual speed of the person is 12 km/hr

Hence, Option A is correct.

42. **B**

We are given that the bus travels at an average speed of 60 km/h when it is moving continuously without any stoppages but including stoppages, its average speed reduces to 45 km/h.

Let's assume the bus travels for 1 hour.

Now, considering the stoppages, the bus will travel 45 km in 1 hour.

Without stoppages, the bus would have covered 45 km in time $= \frac{45}{60} = 0.75$ hours.

Now, the travel time with stoppages is nothing but the sum of the bus's actual travel time and the stoppage time.

$\therefore 1 = \text{stoppage time} + 0.75$

$\text{stoppage time} = 0.25$ hours or 15 min.

This means bus stop for 15 min after running for 45 min.

Thus, the stoppage time is $\frac{1}{3}$ of the running time.

Therefore, for every 1 hour (60 minutes) of actual running:

Stoppage time per hour of running time$= \frac{1}{3} * 60 = 20$ minutes.

Hence, Option B is correct.

43. **D**

We can apply Apollonius theorem here:

$AB^2 + AC^2 = 2\left(AM^2 + BM^2\right)$

$4^2 + 6^2 = 2\left(3^2 + BM^2\right)$

BM=$\sqrt{17}$ cm

Option D.

44. **A**

We know the formula for inradius (r) in terms of Area (A) and semi-perimeter (S).

$r = \frac{A}{S}$

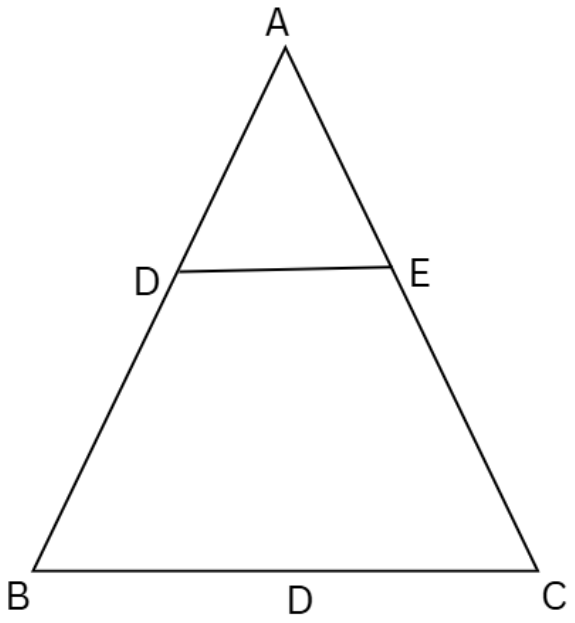
S$= \frac{13 + 14 + 15}{2} = 21$

Using Heron's formula, area = $\sqrt{s\left(s-a\right)\left(s-b\right)\left(s-c\right)}$ = $\sqrt{21\left(21-13\right)\left(21-14\right)\left(21-15\right)}$ = 84

$r = \frac{84}{21} = 4$ cm

Option A

45. **D**



By BPT, we can say that $\frac{AE}{EC} = \frac{AD}{DB} = \frac{1}{2}$

Let the length of AD and AE be x and y respectively.
So, the length of DB and EC is $2x$ and $2y$ respectively.

Now, Area of $\triangle$ADE $= \frac{1}{2} \times x \times y \times \sin A = 9$

So, $xy \sin A = 18$

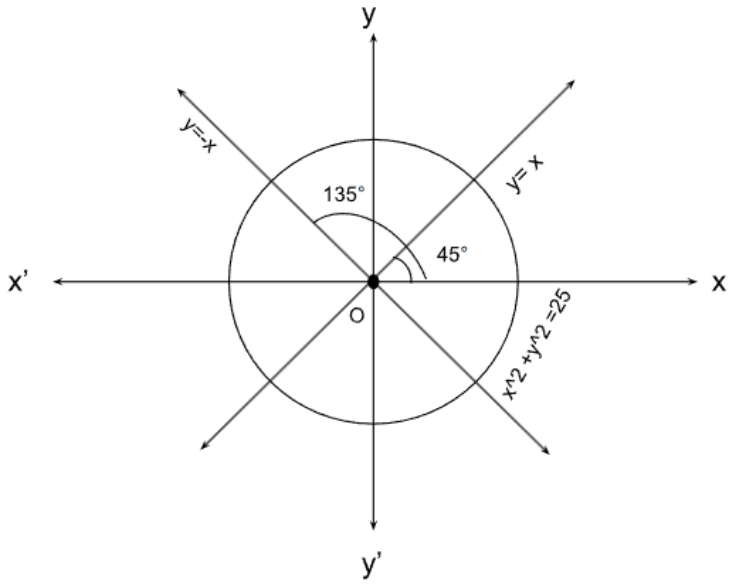
Now, Area of $\triangle$ABC $= \frac{1}{2} \times 3x \times 3y \times \sin A = 81$
Area of trapezium BCED = Area of $\triangle$ABC - Area of $\triangle$ADE $= 81 - 9 = 72$ sq. cm
Option D.

46. **B**
We are given the equations: $x^2 + y^2 \leq 25, y \geq x$ and

$y \geq -x$.

So, let's plot all these equations $x^2 + y^2 = 25, y = x$ and $y = -x$.

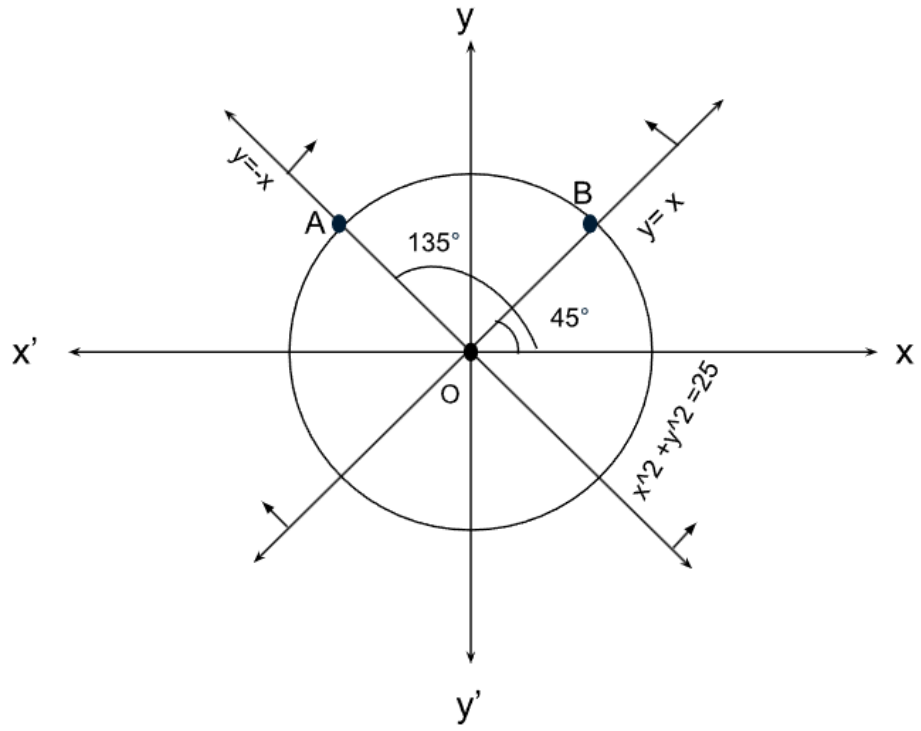
Slope of the lines $y = x$ and $y = -x$ is $45°$ and $135°$respectively.



Now, inequality $x^2 + y^2 \leq 25$ represents all the points inside or on the boundary of the circle $x^2 + y^2 = 25$.

Similarly, inequality $y \geq x$ represents all the points that lie on $y = x$ or towards the left of the line $y = x$.

and, inequality $y \geq -x$ represents all the points that lie on $y = -x$ or towards the right of the line $y = -x$.



Now, the enclosed area is the area of minor sector AOB.

Angle AOB$= 135^\circ - 45^\circ = 90^\circ$

Area of enclosed region or area of minor sector AOB is

$= \frac{90}{360} * (\pi * 5 * 5) = \frac{25\pi}{4}$

Hence, Option B is correct.

47. **10**

Sum of the first n even numbers = $n(n + 1)$

Sum of the first 10 even numbers = $10 \times (10 + 1) = 110$

Sum of the first n odd numbers = n^2

Sum of the first 10 odd numbers = $10^2 = 100$

Difference = $110 - 100 = 10$

Hence, the absolute difference between the sum of the first 10 even numbers and the sum of the first 10 odd numbers is 10.

48. **B**

The given equation is $9x + 2y = 9 - 3xy$

$3xy + 9x + 2y = 9$

In this equation, if we look at the first two terms, we can take $3x$ common and we will get $(y + 3)$.

So, if we can write one more term as $(y + 3)$, then, we can take $(y + 3)$ as common from the LHS.

Add 6 on both sides.

$3xy + 9x + 2y + 6 = 9 + 6$

Take $3x$ common from the first two terms and 2 common from the last two terms.

$3x\,(y + 3) + 2\,(y + 3) = 15$

$(3x + 2) \times (y + 3) = 15$

15 on the RHS has to be written in the form of product of two numbers (as we have a product of two numbers on LHS)

$15 = 1 \times 15, 3 \times 5, -1 \times -15, -3 \times -5$

Now, $(3x + 2)$ can be either one of 5, -5, 3, -3, 15, -15, 1, -1.

So, we get the value of x as $1, \frac{-7}{3}, \frac{1}{3}, \frac{-5}{3}, \frac{13}{3}, \frac{-17}{3}, \frac{-1}{3}, -1$.

Since, it is given that the solution is a non-negative integer,

We get the value of x as 1.

Hence, $(3x + 2)$ = $(3 \times 1) + 2$ = 5

So, we get the value of $(y + 3)$ as 3.

$y = (3 - 3) = 0$

Hence, $x = 1$ and $y = 0$.

$(x \times y) = (1 \times 0) = 0$

Option B.

49. **C**

The given equation is $4xy + 4x + y = 26$

In this equation, if we look at the first two terms, we can take $4x$ common and we will get $(y + 1)$.

So, if we can write one more term as $(y + 1)$, then, we can take $(y + 1)$ as common.

Add 1 on both sides.

$4xy + 4x + y + 1 = 26 + 1$

Take $4x$ common from the first two terms and 1 common from the next two terms.

$4x\,(y + 1) + 1\,(y + 1) = 27$

$(4x + 1) \times (y + 1) = 27$

27 on the RHS has to be written in the form of product of two numbers (as we have a product of two numbers on LHS)

$27 = 1 \times 27, 3 \times 9, -1 \times -27, -3 \times -9$

Now, $(4x + 1)$ can be either one of $9, -9, 3, -3, 27, -27, 1, -1$.

So, we get the value of x as $2, \frac{-10}{4}, \frac{2}{4}, -1, \frac{26}{4}, -7, 0, \frac{-2}{4}$.

Since, it is given that the solution is a positive integer,

We get the value of x as 2.

Hence, $(4x + 1)$ = $(4 \times 2) + 1$ = 9

So, we get the value of $(y + 1)$ as 3.

$y = (3 - 1) = 2$

Hence, $x = 2$ and $y = 2$.

$(x + y) = (2 + 2) = 4$

50. **C**

Speed of the boat in still water (B) = 10 km/hr

Speed of the stream/river current (S) = 2 km/hr

$\Rightarrow$ Downstream speed of the boat = B + S = 10 + 2 = 12 km/hr

The boat covers the downstream distance in 2 hours. Let the distance be D.

$\Rightarrow$ D = 12 $\times$ 2 = 24 km

Now the boat has to travel double the distance upstream.

$\Rightarrow$ New distance = 2D = 2 $\times$ 24 = 48 km

$\Rightarrow$ Upstream speed of the boat = B - S = 10 - 2 = 8 km/hr

$\Rightarrow$ Time taken = 48 / 8 = 6 hours

Hence, option C is the correct choice.

51. **B**

Let N be the total visible steps when the escalator is off.
Let p be the person's speed (steps/sec).
Let e be the escalator's speed (steps/sec).
Let t be the time taken.

When the escalator moves upward and the person walks upward, both move in the same direction: $N = (p + e)t$

Given time (=20) sec, $N = (p + e) \cdot 20 \quad ...(1)$

When the person stands still, only the escalator moves: $N = et$

Given time (=60) sec, $N = e \cdot 60 \quad ...(2)$

From (2), $e = \frac{N}{60}$

Substitute in (1): $N = \left(p + \frac{N}{60}\right) 20$

or, $\frac{N}{20} = p + \frac{N}{60}$

or, $p = \frac{N}{20} - \frac{N}{60} = \frac{3N - N}{60} = \frac{N}{30}$

Now the escalator moves downward while the person walks upward.
So they move in opposite directions: $N = (p - e)t$

or, $N = \left(\frac{N}{30} - \frac{N}{60}\right) t$

or, $N = \frac{N}{60} t;\ t = 60$

So the required time is **60 seconds**.

52. **160**

Since the ratios are preserved, the distance between the current meeting point to B would also increase by 20% if the track length increases by 20%. Thus, the distance between the meeting point to be should be $\frac{840}{1.2} = 700$ meters. This point is in the ratio 1:7 away from A and B respectively. Thus, we get that, $7k = 700$, and hence $(1 + 7)k = 800$, is the total length AB.

Let the speeds of the two runners be x and $7x$ meters/min respectively. If they run in opposite directions, the time taken for them to meet will be

$\Rightarrow \frac{800}{x + 7x} = \frac{100}{x}$ minutes. This is provided as $\frac{100}{x} = 5$ giving us $x = 20$ meters/min.

If the speed of Shyam was twice of what it is now, the two speeds would've been x and $14x$, giving us the meeting time,

$\frac{800}{15x} = \frac{160}{3x} = \frac{160}{20 \times 3} = \frac{8}{3}$ minutes, or $\frac{8}{3} \times 60 = 160$ seconds.

53. **B**

Let the weights of the three pieces be x, $2x$, and $3x$.
The original total weight of the gemstone was $x + 2x + 3x = 6x$.

The value formula is $V = k \cdot (\text{Weight})^2$.

Original Value: $V_{\text{original}} = k \cdot (6x)^2 = 36kx^2$

We are given that the original value is Rs. 72,000:

$36kx^2 = 72000 \Rightarrow kx^2 = 2000$

Combined value of the three broken pieces:
$V_{\text{piece1}} = k(x)^2 = kx^2$
$V_{\text{piece2}} = k(2x)^2 = 4kx^2$
$V_{\text{piece3}} = k(3x)^2 = 9kx^2$
$\text{Total New Value} = kx^2 + 4kx^2 + 9kx^2 = 14kx^2$

$\text{Loss} = 36kx^2 - 14kx^2 = 22kx^2$

Since we know $kx^2 = 2000$:

$\text{Loss} = 22 \times 2000 = 44,000$

54. **C**

The joint variation relationship: $Y = k \cdot F \cdot \sqrt{R}$

First, find the constant of proportionality (k) using the initial conditions ($Y = 2400$, $F = 50$, $R = 64$):

$2400 = k \cdot 50 \cdot \sqrt{64}$
$2400 = k \cdot 50 \cdot 8$

$2400 = 400k \Rightarrow k = 6$

Now, use the formula $Y = 6 \cdot F \cdot \sqrt{R}$ to find the new fertilizer requirement ($Y = 3240$, $R = 81$):

$3240 = 6 \cdot F \cdot \sqrt{81}$

$3240 = 6 \cdot F \cdot 9$

$3240 = 54F \Rightarrow F = \frac{3240}{54} = 60 \text{ kg}$

55. **B**

Let the fixed part for a group be F and the varying constant be k for a particular number of tourists n in a group, such that the total price for the group becomes $F + kn^2$

We have,

$F + 64k = 516000$ ---------(1) and

$(F + 100k) + (F + 4k) = 876000$ or $2F + 104k = 876000$ ----------(2)

The first equation can also be written as $2F + 128k = 1032000$ ----------(3)

Subtracting equation (2) from equation (3), we get

$24k = 156000$ or $k = 6500$. Which gives $F = 100000$

Thus, for the total 10 friends which we have to divide into two groups, we have,

$100000 + 6500n^2 + 100000 + 6500(12 - n)^2$ as the total amount, this gives

$200000 + 6500(2n^2 + 144 - 24n)$ or $200000 + 13000(n^2 - 12n + 72)$

We need to minimise $n^2 - 12n + 72$

The minimum value of this quadratic will occur at $-\frac{(-12)}{2} = 6$, where $n = 6$ gives the minimum cost as;

$200000 + 13000(36 - 72 + 72) = 668000$

The correct answer is Rs. 6,68,000.

56. **A**

$f(x) = kx \ \Rightarrow \ x^2 + px + q = kx \ \Rightarrow \ x^2 + (p-k)x + q = 0$

Now, for the real roots, $D \geq 0$

So, $(p-k)^2 - 4q \geq 0$

So, this must be true for every real value of k.

So, $(p-k)^2 - 4q \geq 0 \ \forall \ \ k \in R$

$(p-k)^2 \geq 4q \ \forall \ k$

$(p-k)^2$ is minimum when $k = p$

So, the minimum value = 0

So,$0 \geq 4q \Rightarrow q \leq 0$

Hence, Option A is correct.

Alternative Method:

$x^2 + (p-k)x + q = 0$

$D_x = (p-k)^2 - 4q \geq 0$

$(p^2 - 2pk + k^2) - 4q \geq 0$

$k^2 - 2pk + (p^2 - 4q) \geq 0$

For a quadratic $Ak^2 + Bk + C$ to be ≥ 0 for all real k:

The leading coefficient must be positive, and the quadratic must have at most one real root. This means its discriminant (D_k) must be less than or equal to zero.

Here: $A = 1, B = -2p, C = p^2 - 4q$:

$D_k = B^2 - 4AC \leq 0$

$(-2p)^2 - 4(1)(p^2 - 4q) \leq 0$

$4p^2 - 4p^2 + 16q \leq 0$

$16q \leq 0$

$q \leq 0$

57. **B**

Let the length of the tunnel be d meters.

The time it takes for train A's last carriage to pass the tunnel after the engine has already passed is $=$ $\frac{\text{length of train A - length of tunnel}}{\text{speed of train A}}$ seconds

The speed of train A in metres per second is $\frac{5}{18} \times 108 = 30$ metres per second.

Thus, we have, $x = \frac{200 - d}{30}$ seconds

Similarly, the speed of train B in metres per second is $\frac{5}{18} \times 90 = 25$ metres per second.

Thus, we further have, $y = \frac{250 - d}{25}$ seconds

We are provided that $x + y = 5\frac{2}{3}$ or $x + y = \frac{17}{3}$

Therefore, we get,

$\frac{200 - d}{30} + \frac{250 - d}{25} = \frac{17}{3}$

$\Rightarrow 5000 - 25d + 7500 - 30d = 4250$

$\Rightarrow 8250 = 55d$ or $d = 150$ metres

58. **D**

Let the price of each unit of Mango be Rs X, and the number of mangoes be N.

Thus, the cost price will be $X \times N = NX$

He later discovers that 15% of the mangoes are rotten and increases the selling price of the remaining mangoes by Rs. 10 per unit. Thus, now he can sell only $$0.85\times N$ mangoes, and the selling price of each mango is $$X+10$$.

Thus, the selling price will be $(X + 10) \times 0.85N$

The wholesaler's overall profit is 19%.

$$19 = \frac{0.85N \times (X + 10) - NX}{NX} \times 100$$

$$0.19X = 0.85 \times (X + 10) - X$$

$$0.19X = 8.5 - 0.15X$$

$$0.34X = 8.5$$

$$X = 25$$

Thus, the Cost Price for the Wholesaler $= NX = N \times 25 = 25N$

Now, 20% of the mangoes were rotten, and he increased the selling price of the remaining mangoes by Rs.15 per unit. Thus, the new selling price will be $= (25 + 15) \times 0.8N = 32N$

Profit %age $= \frac{32N - 25N}{25N} \times 100 = 28\%$

59. **7**

We have $x^2 - x - 1 = 0$, which gives $x^2 = x + 1$ and $x^4 = x^2 + 2x + 1$

Again substituting $x^2 = x + 1$, we get $x^4 = 3x + 2$

We have $\alpha^4 = 3\alpha + 2$, and $\beta^4 = 3\beta + 2$

Giving $\alpha^4 + \beta^4 = 3(\alpha + \beta) + 4 = 3(1) + 4 = 7$

Alternate Solution:

We know that $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \frac{b^2 - 2ac}{a^2} = 3$

Further, $\alpha^4 + \beta^4 = (\alpha^2 + \beta^2)^2 - 2(\alpha\beta)^2$

Therefore, $\alpha^4 + \beta^4 = 9 - 2(-1)^2 = 7$

60. **6**

We can expand the inequality $\left|\left|x^2 - 5x + 2\right| + 7\right| \leq 11$ as,

$-11 \leq \left|x^2 - 5x + 2\right| + 7 \leq 11$

Subtracting -7 from all the equations,

$-18 \leq \left|x^2 - 5x + 2\right| \leq 4$

This expression is in modulus, and it is always going to be positive, so essentially the inequality is,

$0 \leq \left|x^2 - 5x + 2\right| \leq 4$

The left inequality always holds true, focusing on the inequality on the right, it can be written as,

$-4 \leq x^2 - 5x + 2 \leq 4$

Adding +4 to all the expressions,

$0 \leq x^2 - 5x + 6 \leq 8$

Now we are just looking at integral numbers that satisfy the above inequality,

First part of the inequality is,

$0 \leq x^2 - 5x + 6$

$0 \leq (x - 3)(x - 2)$

This is true for all integers as the numbers for which the above expression is negative lies between (2,3).

Secondly,

$(x - 3)(x - 2) \leq 8$

This is true for 4, 5

For 6, the value becomes 12 and since this is an increasing function, it will only increase.

Thus the number of non-negative integers that satisfy the inequality are,

0, 1, 2, 3, 4, 5

6 of them satisfy

61. **C**

The given inequality is $|X+1||X+7|<12$

We can check for some values and see the pattern for the values of x that are possible.

When X = 0, 1*7 < 12

X = 1, 2*8 > 12 (Hence, not possible)

X = -1, 0< 12

X = -2, 5< 12

X = -3, 8<12

.

.

X = -7, 0< 12

X = -8, 7< 12

X = -9, 16>12

Hence, the only values that satisfy are X = 0,-1,-2,-3,-4,-5,-6,-7,-8.

Thus, there are 9 possible values.

62. **C**

$$\frac{|x-2|\left(x^2-11x+28\right)}{\left(x^2-2x-35\right)}<0$$

The quadratic in the denominator can be factorized as $x^2-2x-35=(x+5)(x-7)$

The quadratic is positive for $x>7$ and $x<-5$.

The numerator has a modulus i.e. the value of $|x-2|$ is always non-negative, so it won't effect the sign of the inequality.

But $x\neq\ 2$ for the inequality to be satisfied.

The quadratic in the numerator can be factorized as $x^2-11x+28=(x-4)(x-7)$

The quadratic is positive for $x>7$ and $x<4$.

We can cancel out the $x-7$ from both the numerator and denominator.

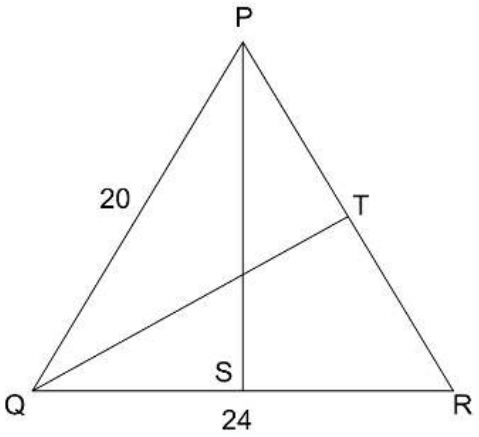
We get $\frac{|x-2|(x-4)}{(x+5)}<0$

For the inequality to be satisfied, both the numerator and denominator should have opposite signs i.e. $x\in(-5,\ 4)-\{2\}$

So, the integral values are {-4, -3, -2, -1, 0, 1, 3} i.e. a total of 7 values

Hence, the answer is 7.

63. **A**



From the figure, we have the length of side PQ, QR and median PS of length $8\sqrt{7}$ cm.

We know QS = QR/2 = 12 cm

Using Appolonius theorem, we can find the length of side PR

$PQ^2 + PR^2 = 2\left(PS^2 + QS^2\right)$

$20^2 + PR^2 = 2\left(\left(8\sqrt{7}\right)^2 + 12^2\right)$

$400 + PR^2 = 2\left(448 + 144\right)$

$PR^2 = 784$

$PR = 28$

Now as PR = 28 cm, RT = PR/2 = 14 cm

Using Appolonius theorem again we get

$QR^2 + QP^2 = 2\left(QT^2 + RT^2\right)$

$24^2 + 20^2 = 2\left(QT^2 + 14^2\right)$

$576 + 400 = 2\left(QT^2 + 196\right)$

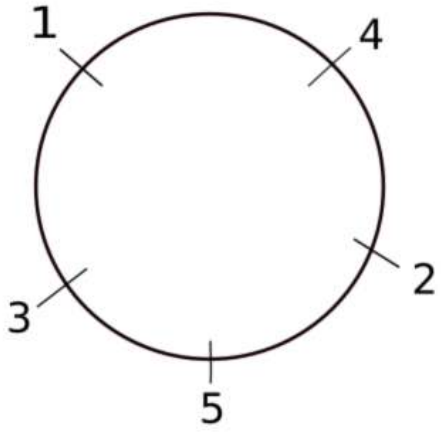
$QT^2 = 292$

$QT = 2\sqrt{73}$

64. **B**

As the speeds of Ajay and Vijay is in the ratio 2 : 3 and they are running in the opposite directions, the total number of meeting points will be 5.

The 9th meeting point will be the same as the 4th meeting point.



Assuming that Ajay and Vijay start from 5 and Ajay is travelling clockwise while Vijay is travelling anti-clockwise

1st meeting will be at 1.

2nd meeting will be at 2.

3rd meeting will be at 3.

4th meeting will be at 4

5th meeting will be at the starting point i.e. 5 and the cycle repeats.

So, 9th meeting will be at 4, which is given that it is 200 m from the starting point.

If the distance between 2 consecutive meeting points is x

So, $2x = 200$

or, $x = 100$

Now, for each meeting Ajay travels 200 m and Vijay travels 300 m

So, for the 7th meeting point Ajay would travel $7 \times 200 = 1400$ m

65. **D**

The surface area of 2m x 2m x 2m cube is $6a^2$ or $6a^2 = 6 \times (2)^2 = 24\ m^2$

As the profit is proportional to the surface area, assume the constant of proportionality to be k

So, $24k = 10000$

or $k = \dfrac{10000}{24} = \dfrac{1250}{3}$

Now, the combined area of the five blocks of wood is calculated as,

The surface area of cuboid is $2\,(lb + bh + hl)$

i.e. $2\,(lb + bh + hl) = 2\,(4 + 2 + 2) = 16\ m^2$

The surface area of the four cubes is $4 \times \left(6 \times 1^2\right) = 24$

So, the combined area now is 16+24 = 40

Hence, the profit now will be $40k = 40 \times \dfrac{1250}{3} = 16667$

Hence, the answer is Rs. 16667

66. **C**

Let's take the work done my one man in a single day to be 20 units.
This would give the work done by a woman and a child per day to be $\left(20 \times \frac{4}{5}\right) = 16$ and $\left(20 \times \frac{1}{2}\right) = 10$ units respectively.

The total work then would be $55 \times 6 \times 20$ = 6600 units

The approach should be to find the lowest and highest element of set S.
All the values in between those two (integer number of days) would be possible for some combination of women and children hired on particular days.

The lowest time will be taken when everyday a woman is hired.
the work done will be $16 + (16 \times 2) + (16 \times 3) + ... (16 \times n)$ this has to be greater than equal to 6600
Giving the equation:
$\frac{16n(n+1)}{2} \geq 6600$
$n\,(n+1) \geq 825$

{A quick way of solving such equations is to find the square root of 825, that would be a close value to n}
Using square root of 825, we can guess that n would be around 28.7
Putting n=28 we get work done to be 812, and putting n=29 we get work done to be 870
Hence, if all the women are hired, it would take the minimum time which is 29 days.

The longest time would be taken when all the children are hired.
That would give us the inequality:
$\frac{n(n+1)}{2}10 \geq 6600$
$n\left(n+1\right) \geq 1320$
Using similar method, we can get the value of n to be 36 days.

So the task can be competed in a minimum of 29 and a maximum of 36 days.
All the days in between, 30-35 would be possible for some combination of woman and children hired on some particular day.

Therefore, the sum of all elements of S would be 260.

Therefore, Option C is the correct answer.

67. **D**

Let us assume that the interest rate is r. And the principal amount is P.

The sum after three years would be, $P\left(1+\frac{r}{100}\right)^3$, we are told that this value is 13380.

Similarly, the sum after six years would be, $P\left(1+\frac{r}{100}\right)^6$, we are told that this value is 20,070.

Let us divide the latter by the former, we get,

$\frac{P\left(1+\frac{r}{100}\right)^6}{P\left(1+\frac{r}{100}\right)^3} = \frac{20070}{13380} = 1.5$

Therefore, $\left(1+\frac{r}{100}\right)^3 = 1.5$

Substituting this value in the first equation we have, $P\left(1.5\right) = 13380$

We get P = 8920.

Hence, the answer is Option D.

68. **A**

It is given,

$\frac{P\times 8\times 13}{100} = 6500$

P = Rs 6250

Amount after 2 years = $6250\left(1+\frac{8}{100}\right)^2 = 6250\times\frac{108}{100}\times\frac{108}{100} = 7290$

Compound interest = 7290 - 6250 = Rs 1040

The answer is option A.

69. **D**

Let the speed of Himanshu be X metre/ minute, that of Nikhil be Y metre/minute, and the length of the circular track is L meter.

Then, according to the question, when they walk in the same direction, Nikhil completes 3 more rounds than Himanshu, which means the speed of Nikhil (Y) is more than Himanshu's speed (X)

=> 45(Y-X) = 3L -------equation(1)

If they walk in opposite directions, they meet after 5 minutes

=> 5(Y+X) = L --------equation(2)

Solving both equations simultaneously, we get,

45Y-45X = 3L

45Y+45X = 9L

=> 90Y = 12L, Y= $\frac{2L}{15}$

Then, X = $\frac{L}{15}$

The speed ratios = 1:2

70. **41**

Let the time spent by Kushal on the paper be x. The time spent by Aman will be 3x.

It is given that the total time spent by both was 36 hours.

Thus, x + 3x = 36

=> x = 9 hours

Now, let the paper written by Aman be y per hour.

Speed of Kushal will be y + 4.

Thus, 9(y + 4) + 27y = 288

=> 36y + 36 = 288

=> 36y = 252

=> y = 7.

If the speeds are interchanged, Kushal will write 7 pages per hour.

Thus, the time taken by Kushal = $\frac{288}{7} \approx 41$

71. **80**

Since time is constant, the ratio of speeds is the ratio of the distance.

Let the speeds of X, Y and Z be x, y and z km/hr.

x:y:z = circumference of A:circumference of B:circumference of C = 18:7:10

x:y:z = 18:7:10

x = 18k

y = 7k

z = 10k

18k = 144

k = 8 km/hr

y = 56 km/hr

z = 80 km/hr

z - y = 80 - 56 = 24 km/hr = 24 x 5/18 m/s

r = 20/3

12r = 80.

72. **B**

Let the cost of milk be Rs. 100/L and the cost of water be Rs. 1/L.

Hence, CP = 200 + 3 = 203.

SP = 5 X 100 = 500.

Profit % = (500 - 203)/203 x 100 = 297/203 x 100 = 146%(approx)

73. **A**

Given that we are unknown to the ages of arjun and atul .

Let them be x and y .

Given (x +1/y)(y +1/x) = 49/6

xy +1/xy = 37/6

Solving this we get xy = 6 .

Now we know that none of them are 1 year .

Hence we can have x , y to be 2 and 3 in any order .

74. **A**

Let the markup be x%, And the original price be P

As per the question

$P\left(1+x\%\right)\left(1-\frac{x}{2}\%\right)=1.08P$

Thus $\left(1+\frac{x}{100}\right)\left(1-\frac{x}{200}\right)=1.08$

Upon Expanding we get

$1+\frac{x}{100}-\frac{x}{200}-\frac{x^2}{20000}=1.08$

$\frac{x}{200}-\frac{x^2}{20000}=0.08$

Multiply by 20000 both sides

$100x-x^2=1600$

$x^2-100x+1600=0$

$x^2-80x-20x+1600=0$

Thus $(x-20)(x-80)$= 0

Thus x = 20 or 80

Option is the only correct option

75. **B**

Let Ram current age be r and Shyam's be s

$\frac{r+10}{s+10}=\frac{6}{7}$

$7r+10\ =\ 6s$ .....(I)

Their age differece will always remain same thus s-r = 5 ....(II)

Putting eqn value of s from eqn(II) to eqn (I)

$7r+10\ =\ 6(r+5)$

$7r+10\ =\ 6r+30$

On solving we get r =20

76. **D**

5200 mangoes were left in the whole farm.

Had he plucked 20 more mangoes from each tree, 4400 mangoes would have left in the farm.

5200-4400=800

Number of trees = 800/20=40 trees

Total number of mangoes in the farm initilally= 240*40=9600

x mangoes from each tree

9600-40x= 5200

x= 4400/40= 110

Option D

77. **D**

Consider the number of first-class tickets =a and the second-class ticket =b

10a+3b=110

a+b=18

Solving two equations, we get a=8, b=10

After interchanging the number of tickets, the new cost = 10*10+3*8 =124

78. **C**

Let us assume the number of apples purchased by the man to be x.

Then, the number of oranges will be (40-x).

Let the price of each apple be Rs. y and that of an orange be Rs. z

It is given that, xy+ (40-x)z= 17.

=> xy+40z-xz=17 ----------(1)

Now, if the number of apples and oranges are interchanged, the apples will be (40-x) and the oranges will be x.

Given, the new total cost will be Rs. 15

So, (40-x)y + xz=15

=>40y-xy +xz= 15 ----------(2)

Adding eqns 1 and 2, we get,

xy+40z-xz+40y-xy+xz=15+17

=> 40(y+z)=32

=>y+z= $\frac{32}{40} = \frac{4}{5}$

Or, the cost of one apple and one orange, x+y= Rs. $\frac{4}{5} = 80\ paise$

79. **B**

Let the number of overtime hours logged be x.

Let the number of accidents reported be A.

So, since the relationship between A and x is linear, we can write A= ax+c

Given, when x=1000, A=8.

So, 8= 1000a+ c----------(1)

Again, when x= 400, A=5.

So, 5= 400a+ c ----------(2)

Subtracting Eqn 2 from 1, we get,

600a= 3

=> a= 1/200.

Now putting this value of a in Eqn 1, we find

8= $\frac{1000}{200} + c$

=>c= 8-5=3.

So, when x=0,

A= c= 3- option B

80. **D**

Let x, y, z be the number of items purchased by Nisha of price Rs. 5, 2, 1 respectively.

5x+2y+z = 20

Since the shopkeeper had no change with him/her; he/she gave Nisha three more products of price Rs. 1 each i.e Nisha purchased items of Rs. 17

5x+2y+z =17

She purchased all three types of products in more than one quantity,

If she purchases three items of Rs. 5, she cannot purchase 2 items of Rs 2 and 1

So Nisha purchases 2 items of Rs. 5.

If Nisha purchases 3 items of Rs 2., then she cannot purchase more than 1 item of Rs. 1.

Hence she must have purchased 2 items of Rs. 2 and 3 items of Rs. 1.

The number of items Nisha will have at the end of the transaction = 2+2+3+3(due to no change)

= 10

D is the correct answer.

81. **C**

Let the filling capacity of the pump be x m^3/min

Then the emptying capacity of the pump = (x+10) m^3/min

The time required for filling the tank = $\frac{2400}{x}$

The time required for emptying the tank = $\frac{2400}{x+10}$

The pump needs 8 min less to empty the tank than to fill it.

$\frac{2400}{x}$ - $\frac{2400}{x+10}$ = 8

$\frac{300}{x}$ - $\frac{300}{x+10}$ = 1

300(x+10) - 300(x) = x(x+10)

$x^2 + 10x - 3000$ = 0

(x+60)(x-50) = 0

x = 50 or -60

Since x cannot be negative, x = 50

Filling capacity of the pump = 50 m^3/min

C is the correct answer.

82. **A**

1st case :

Distance travelled by train = 300km and by taxi = 200km

Let the speed of train be x km/hr and that of taxi be y km/hr.

Time taken by train = 300/x and time taken by taxi = 200/y hr

$\frac{300}{x}$ = $\frac{200}{y}$

Total time taken = 5hrs 30 mins = 5 1/2 hrs

$\frac{300}{x}$ + $\frac{200}{y}$ = 5 $\frac{1}{2}$

On solving we get : 600y + 400x = 11xy ----- Eq (1)

2nd case :

Distance travelled by train = 260 km and by taxi = 240 km

The speed of train and taxi will remain the same.

Time taken by train = $\frac{260}{x}$ hr and by taxi = $\frac{240}{y}$ hr

$\frac{260}{x}$ + $\frac{240}{y}$ = 5hrs 36 mins

$\frac{260}{x}$ + $\frac{240}{y}$ = 5 $\frac{3}{5}$ on solving we get,

1300y + 1200x = 28xy ---- Eq (2)

Solving 1 and 2 we get:

x=100 km/hr

Speed of the train = 100km/hr

A is the correct answer.

83. **B**

Let 100x be the total numbers of voters in the voters list. => 90x voters voted

A=1.4B (since A received 40% more votes than B)

C=1.2B

Given A-C = 4000

1.4B-1.2B=4000 => B=20000 => A=24000 and C = 28000

So total voter who voted =A+B+C= 72000 which is 90% of the total list.

So when 90%= 72000 , 100%= 80000.

84. **C**

At end of the game each player holds = 32

Total money is 4×32=Rs. 128

Let Frank start with Rs. x

Then the total money other 3 have =Rs. (128−x)

Frank lost the first game. He gives Rs. (128−x) to the other 3 players.

Therefore, after the 1st game, money that Frank has

=x−(128−x)=2x−128

Total money other 3 players have =2(128−x)

In the second, third, and fourth games, the other 3 partners will lose, and each time, the loser doubles the amount of money that the others have.

i.e., after 2nd game, Frank money =2(2x−128)

i.e., after 3rd game, Frank money =4(2x−128)

i.e., after the 4th game, Frank money =8(2x−128)

8(2x−128) = 32

$x = 66$

At the start of the game, Frank started with 66rs.

Alternate solution:

Since, all the players lost in the reverse alphabetical order, below table gives the result of which player lost in each round.

	Frank	Faulad	Farhan	Fardeen
Round 1	Lost			
Round 2		Lost		
Round 3			Lost	
Round 4				Lost

Now, each player had ₹32 at the end of Round 4. It means that the total money with the friends in the game is $₹32 \times 4 = ₹128$

Now, Fardeen lost in round 4. It means that Fardeen doubles the amount of money of others.

Total money with Frank, Faulad and Farhan after Round 4 = $₹32 \times 3 = ₹96$

Total money with Frank, Faulad and Farhan after Round 3 = $\frac{₹96}{2} = ₹48$

Money with Fradeen after Round 3 = $₹128 - ₹48 = ₹80$

Now, Farhan lost in Round 3.

Total money with other 3 friends after Round 2 = $₹\left(8 + 8 + 40\right) = ₹56$

Total money with Farhan after Round 2 = $₹\left(128 - 56\right) = ₹72$

Similarly, Faulad lost in Round 2.

Total money with other 3 friends after Round 1 = ₹ $(4 + 36 + 20) = ₹60$

Total money with Faulad after Round 1 = ₹ $(128 - 60) = ₹68$

Similarly, Frank lost in Round 1.

Total money with other 3 friends at the start = ₹ $(34 + 18 + 10) = ₹62$

Total money with Frank at the start = ₹ $(128 - 62) = ₹66$

The total money with each of the friend after each round is:

	Frank	Faulad	Farhan	Fardeen
Round 0	66	34	18	10
Round 1	4	68	36	20
Round 2	8	8	72	40
Round 3	16	16	16	80
Round 4	32	32	32	32

Hence, Frank starts with the most money at the beginning of the game.

$\therefore$ The required answer is C.

85. **A**

$f(x) = ax^2 + bx + c <$ x

$ax^2 + x\,(b - 1) + c\ < 0$

Here, x^2 is the significant term that influences the value of the expression. So, if the coefficient of this term is positive, irrespective of the other terms, the expression will be greater than x when the value of x goes higher.

But if the value of a is negative, then this curve will be an inverted U-curve. So, $a \leq\ 0$ is a correct condition.

86. **D**

When two people run on a circular track with speeds a m/s and b m/s in the same direction, then the number of distinct meeting points will be |x - y| where x:y is a:b in the minimal format.

If they meet at the starting point the fourth time they meet, then they can meet at 1 distinct point or 2 distinct points or 4 distinct points. This corresponds to |x-y| being 1, 2 or 4.
Now let us take a look at the options.
Option A - x=3 and y=4. |x-y| = 1. Hence, this option is valid.
Option B - a:b = 36/28 = 9/7. Hence, |x-y| = 2. Hence, this is a valid option.
Option C - a:b = 35/15 = 7/3. Hence, |x-y| = 4. Hence, this is a valid option.
Option D - a:b = 28/7 = 4/1. Hence, |x-y| = 3. This is not a valid value. Thus, the answer is option D.

87. **D**

$3x^2 - 4x + 7$ is the given quadratic equation

The smallest value of a given quadratic equation $ax^2 + bx + c = 0$ occurs at $x = \dfrac{-b}{2a}$

Thus, the smallest value of the given expression will occur at $x = \dfrac{-(-4)}{2 * 3} = \dfrac{2}{3}$

Thus, the smallest vale of the given expression is $3 * (\frac{2}{3})^2 - 4 * (\frac{2}{3}) + 7 = \frac{17}{3}$

Hence, option D is the correct answer.

88. **C**

Let x and y be the percentage of aluminum in the 8 kg and 16 kg piece.
Let us take a kg away from both.
Thus, we will have xa and ya amount of aluminum in both the pieces.
In the remaining 8-a and 16-a pieces we will have (8-a)x and (16-a)y amount of aluminum.
Given that, $\frac{(8-a)x + ay}{8} = \frac{(16-a)y + ax}{16}$
Thus, $2 * (8x - ax + ay) = 16y - ay + ax$
=> $16x - 2ax + 2ay = 16y - ay + ax$
Thus, $16(x - y) = 3a(x - y)$
Thus, $a = \frac{16}{3} = 5.33$
Hence, option C is the correct answer.

89. **B**

Sukriti covers a distance of 1 km in 5 minutes and 50 seconds therefore will cover 5 kms in 25 mins 250 secs = 29.16 mins

Saloni covers the 1 km in 6 minutes and 4 seconds therefore her speed is $\frac{1}{6.06}$ kms per minute

Therefore Distance Covered by Saloni in 29.16 mins is 29.16 mins* $\frac{1}{6.06} \approx$ 4.8 kms

Therefore the distance with which Sukriti will win a 5 km mini marathon is $\approx$ 200 meters

therefore our answer is option 'B'

90. **A**

The bus driver earns Rs. 1080. Therefore, he must have stopped for 1080/12 = 90 minutes.
The bus takes 7 PM - 10 AM = 9 hours to complete the journey. But, the bus actually runs for 9 - 1.5 = 7.5 hours.
This implies that the distance between 2 cities is 7.5 * 54.
The bus completes the entire journey in 9 hours. Let the average speed of bus be ‘x’. Then, distance between the 2 cities will be 9x.

Since the distance is constant, we can equate 9x and 7.5*54.
x = 7.5*6
x = 45 kmph.
Option A is the right answer.

91. **A**

A and B met for the first time at point 7.

Let the speeds of A, B and C be a,b, and c.

A and B meet for the first time at point 7. Hence, by symmetry, they must meet at point 1 again. They will be meeting at only 2 distinct points.

Since they are running in the same direction, the number of distinct points at which they will meet is given by a-b.

a-b = 2

=> (a,b) = (3,2), (4,2), (5,3), and so on. It is given that a/b >2. Only (3,1) satisfies this condition.

Similarly, A and C meet for the first time at point 4. They must meet at points 4, 7,10 and 1 by symmetry.

Hence, a-c = 4. (a,c) can be (5,1) , (6,2), (7,3), (8,4) and so on.

As B is faster than C, only (5,1) satisfies the condition.

=> a:c = 5:1

We have already found out that a:b = 3:1

Therefore, b:c =5:3

They will meet at 5-3 = 2 distinct points. One of them must definitely be the starting point. Hence, by symmetry, the other point must be diametrically opposite to point 1.

Hence, they will meet at point 7.

Option A is the right answer.

92. **B**

Let the side of the equilateral triangle ABC be 'a'
Let the in-radius and circumradius of the triangle ABC be 'r' and 'R' respectively.
Let the side of the square be 's'
Diagonal of the square $\sqrt{2}s = 2r$
=> $s = \sqrt{2}r$
Area of the square = $2r^2$
Area of the circumcircle is πR^2
Required ratio = $\frac{2r^2}{\pi R^2}$ [For an equilateral triangle, r:R = 1:2]
= $\frac{1}{2\pi}$

93. **B**

"Pipe A of square cross section with side 4cm can fill an empty tank in 30 minutes" ==> volume of the tank = $30(60) * 4^2 * x$ ----(1)

"Pipe B can empty the full tank in 20 minutes" ==> Volume of the tank = $20(60) * \pi * 7^2 * y$ ----(2)

by equating both the equations we get 12x = 77y.

Therefore based on the given conditions the values which satisfy X, Y

x = 77cm/sec

y = 12 cm/sec

volume of the tank = $(30(60) * 4^2 * 77) = 2217600\ cm^3$ ==> 2.2176 $m^3 \approx 2.22m^3$

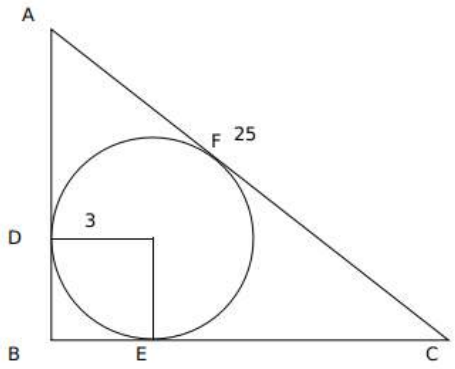
94. **A**

Let BD = x, BE = y, AD = DE = EC = z = 10/3.
Applying Apollonius Theorem in triangle ABE,
$6^2 + y^2 = 2 * (x^2 + z^2)$ ---- (1)
Similarly applying Apollonius Theorem in triangle BDC,
$8^2 + x^2 = 2 * (y^2 + z^2)$ ---- (2)
Adding equations (1) and (2) we get,
$100 + x^2 + y^2 = 2x^2 + 2y^2 + 4z^2$
=> $x^2 + y^2 = 100 - 4z^2$
We know that z = 10/3, so $z^2 = 100/9$
=> $x^2 + y^2 = 100 - \frac{400}{9}$
=> $x^2 + y^2 = \frac{500}{9}$

95. **C**

It is given that the circumradius of the right triangle is 12.5. Thus hypotenuse will be 12.5*2 =25

Since the inradius is 3, Let the triangle be ABC and incircle touches the side at D,E,F. Upon drawing a figure we get the following.



Here DB = BE = 3

Since AD and AF are tangent to the incircle they will be equal hence AD = AF ....(i)

Similarly, EC and FC are equal as they are also tangent from same point thus, CF =CE ..(ii)

AC = 25 = AF+FC

Perimeter = AB+BC+CA = AD+DB+BE+EC+CF+FA

Putting the value we from above we get that perimeter = 3+3+ 2(AF+FC) = 3+3+2(25) = 6+50 = 56

Alternate solution: Let 'l' and 'b' be the height and base of the right-angle triangle.

We know that hypotenuse of a right angle triangle = 2*circumradius.

Hence, we can say that the hypotenuse of the right-angle triangle = 2*12.5 = 25 cm.

Therefore, we can say that $l^2 + b^2 = 25^2$ {Using Pythagoras theorem}

$\Rightarrow (l+b)^2 - 2lb = 25^2$

$\Rightarrow (l+b)^2 = 25^2 + 2lb$ ... (1)

Inradius of a triangle = $\frac{\text{Area of the triange}}{\text{Semi-perimeter}}$

$\Rightarrow 3 = \frac{1/2 * l * b}{\frac{25+l+b}{2}}$

$\Rightarrow 25 + l + b = 1/3 * l * b$

$\Rightarrow l + b = 1/3 * l * b - 25$

$\Rightarrow (l+b)^2 = (1/3 * l * b - 25)^2$ ... (2)

From equation (1) and (2) we can say that

$\Rightarrow 625 + 2lb = \frac{(lb)^2}{9} + 625 - \frac{50lb}{3}$

$\Rightarrow lb = 168$ ... (3)

From equation (1) and (3) we can say that $(l+b)^2 = 961$ or $l + b = 31$

Therefore, the perimeter of the triangle = l+b+h = 31 + 25 = 56 cm.

96. **D**
The ratio of water in the three samples = 1/4, 2/3 and 3/5.
Since the samples are mixed in the ratio 1:2:3, the proportion of water in the final sample = $\frac{1}{6}(\frac{1}{4} + \frac{2}{3} \times 2 + \frac{3}{5} \times 3) = 203/360$
Thus, the final ratio of milk and water = $\frac{203}{360-203}$ = 203/157

97. **D**
Let the original number of students in the class be N.
Total weight of the class = 54N
New total weight of the class = 54N + 145
New average weight of the class = (54N + 145)/(N+1) = (54N + 54)/(N+1) + 91/(N+1) = 54 + 91/(N+1).
Since the new average is an integer, (N+1) should be a factor of 91.
If N+1 = 7, the new average becomes 54 + 91/7 = 54 + 13 = 67
and if N+1 = 13, then the new average becomes 54 + 91/13 = 54 + 7 = 61
Both 67 and 61 are prime numbers less than 72. So, we cannot uniquely determine the number of students in the class.

98. **D**
Let the distance of the meeting point from city X be d.
So, distance travelled by A = d
Speed = 70 kmph
Time taken = d/70
Distance travelled by B = 420 - d
Speed of B = 60 kmph
Time taken = (420-d)/60
So, d/70 = (420-d)/60 + 1
Solving this, we get, d = 258.46 km = 258 km (approx.)
So, A and B met at a distance of 258 km from city X.

99. **A**
Let the amount he is supposed to receive be x Rs and y paise.

He instead received y Rs and x paise. The difference = 99 paise.

$$100x + y - (100y + x) = 99$$

$$99x - 99y = 99$$

$$x - y = 1$$

Implies x is more than y by 1.

This is possible for various bus ticket values. For example, if the bus ticket = 3.95, the amount he was supposed to receive = 6 rupees and 5 paise. (x is more than y by 1). So, this is an acceptable value.

For 7.97, he would receive Rs. 2 and 3 paise

For 5.95, he would receive Rs. 4 and 5 paise

So, only option A is correct.

100. **C**
8 beedis, 11 cigars and 14 cigarettes cost 380 Rs. .... (a)
5 beedis, 7 cigars and 9 cigarettes cost 240 Rs. .....(b)
Subtracting (a) from (b), we get
3 beedis, 4 cigars and 5 cigarettes cost 140 Rs .... (c)

Subtracting (c) from (b), we get
2 beedis, 3 cigars and 4 cigarettes cost 100 Rs. ....(d)
Subtracting (d) from (c) we get
1 beedi, 1 cigar and 1 cigarette cost 40 Rs

101. **B**
20, 99 and 101 form a pythagorean triplet.
Area of the triangle = 1/2 *20*99 = 990
990 = rs
s = (101+99+20)/2 = 110
r*110 = 990
r = 9 cm

102. **80**
In the duration given, the ratio of distances covered by Vijay and Ram would be equal to 3/5
Let the distance covered by Vijay inside the tunnel be y.
Distance covered by Vijay = 120-y+80
Distance covered by Ram = 120+80 = 200
$\frac{200}{200-y} = \frac{5}{3}$
y = 80m

103. **C**
The value of the product is 0 when p is -5 or 15.
p lies between -5 and 15 so 15-p and p+5 are positive(p not equal to -5 or 15)
So, we can apply AM $\geq$ GM here.
$\frac{(\frac{15-p}{9} + \frac{15-p}{9} + ..(9\times)... + \frac{15-p}{9}) + (\frac{p+5}{11} + \frac{p+5}{11} + ...(11\times).... + \frac{p+5}{11})}{20} \geq ((\frac{15-p}{9})^9 \times (\frac{p+5}{11})^{11})^{\frac{1}{20}}$

$\frac{(15-p)+(p+5)}{20} \geq \left(\left(\frac{15-p}{9}\right)^9 \cdot \left(\frac{p+5}{11}\right)^{11}\right)^{\frac{1}{20}}$

$1 \geq \left(\left(\frac{15-p}{9}\right)^9 \cdot \left(\frac{p+5}{11}\right)^{11}\right)^{\frac{1}{20}}$

=> $(15-p)^9 * (p+5)^{11} \leq 11^{11} * 9^9$
Equality holds when $\frac{15-p}{9} = \frac{p+5}{11}$ i.e when p = 6

104. **D**
For summation of square of roots to be zero, individual roots should be zero.
Hence summation should be zero i.e. A-3=0 ; A = 3
And product of roots will also be zero i.e. A-2 = 0 ; A =2
So there is no unique value of A which can satisfy above equation.

105. **D**
The sum of interior and exterior angles is 180. So, the interior angle is 140 and the exterior angle is 40. The exterior angle is also equal to 360/n
So, 360/n = 40 or n = 9